CHAPTER-1

INTRODUCTION

1.1 OVERVIEW

The Student Registration Tracking System is a cloud-based application developed using the Salesforce platform. The project uses Salesforce technologies such as Experience Cloud, Flows, Approval Processes, Reports and Dashboards to manage and track student course registrations. The main aim of the system is to provide a centralized and automated platform that simplifies the process of course management and student registration within the placement department.

In many educational institutions course registration and student participation are often managed manually using spreadsheets, emails or paper. The Student Registration Tracking System helps overcome these issues by providing an automated system that manages the entire process efficiently. In this system the System Administrator is responsible for creating courses and managing the overall system configuration. After a course is created it must go through an approval process where the Placement Head reviews the course and either approves or rejects it.

The only approved courses are made available for students, ensuring that all courses offered are verified and authorized. Once a course is approved the system sends registration emails to students containing a link to the registration page. Students can access the website and register for the course through the Experience Cloud portal without needing to log in. The registration details are stored in Salesforce allowing the system to maintain accurate records of student participation. The system also includes reporting and dashboard features that allow the Placement Head and Placement Faculty to monitor course registrations and student activity.

1.2 EXISTING SYSTEM

In many educational institutions the process of managing student registrations for training or placement courses is usually handled using manual methods. Course details are often shared through emails, notice boards or messaging platforms and students provide their details through forms or simple responses. The placement department then collects and organizes this information manually which can take a significant amount of time and effort.

The existing system commonly depends on spreadsheets or paper-based records to store and track student registration details. Placement staff must manually enter student information, update course details and monitor the number of registrations. This manual approach increases the chances of errors such as duplicate entries, missing information and difficulty in maintaining accurate records especially when a large number of students are involved.

Another limitation of the existing system is the lack of automation and structured approval processes. Courses may be shared with students without proper verification or approval from the placement head. Additionally, there is no automated method to notify students or track registrations efficiently making it difficult for the placement department to monitor participation and manage training activities effectively.

1.3 DRAWBACKS OF EXISTING SYSTEM

- Course registration is handled manually which increases the workload for placement staff.
- Student data is often maintained in spreadsheets or paper records which can lead to data loss.
- Manual data entry increases the chances of human errors and duplicate records.
- There is no proper approval mechanism for verifying courses before sharing them with students.
- Tracking student registrations becomes difficult when the number of students increases.
- Monitoring course participation and student activity is time-consuming.

1.4 PROPOSED SYSTEM

The proposed system is a Student Registration Tracking System developed using the Salesforce platform to automate and manage the process of course registration for students. The system uses Salesforce features such as Experience Cloud, Flows, Approval Processes, Reports and Dashboards to provide a centralized and efficient solution. In this system, the System Administrator creates courses, which are then sent for approval to the Placement Head. The Placement Head reviews the course and either approves or rejects it ensuring that only authorized courses are made available to students.

Once a course is approved the system automatically sends registration emails to students with a course registration link. Students can access the Experience Cloud website and register for the course without needing a Salesforce login. All registration details are stored within the system allowing the Placement Head and Placement Faculty to monitor student participation through reports and dashboards which helps in better tracking and management of training programs.

1.5 ADVANTAGES OF PROPOSED SYSTEM

- The system automates the student course registration process.
- It reduces manual work for the placement department.
- Course approval is properly managed through an automated approval workflow.
- Only approved courses are visible to students for registration.
- Students receive automated email notifications for course registration.
- All student and course data are stored in a centralized system.
- The system reduces errors caused by manual data entry.

SUMMARY

This chapter introduced the Student Registration Tracking System developed using the Salesforce platform and explained the purpose of automating the student course registration process. It discussed the existing manual system and highlighted the difficulties involved in managing registrations without automation. The chapter also described the proposed system that uses Salesforce technologies to improve efficiency. The benefits of implementing an automated registration system were explained. The advantages of the proposed solution over the existing system were also presented.

CHAPTER-2

SYSTEM ANALYSIS

2.1 IDENTIFICATION OF NEED

In many educational institutions, managing student registrations for training or placement-related courses is often done using manual methods such as spreadsheets, emails or paper-based forms. This process makes it difficult for the placement department to efficiently organize course details, track student registrations and maintain accurate records. As the number of students and courses increases managing this information manually becomes time-consuming and prone to errors.

Therefore, there is a need for an automated system that can simplify the course registration and tracking process. The Student Registration Tracking System developed using Salesforce helps address these issues by providing a centralized platform where courses can be created, approved and shared with students efficiently. The system also enables easy monitoring of student registrations through reports and dashboards improving overall management and transparency.

2.2 FEASIBILITY STUDY

A feasibility study is a complete analysis that is conducted to evaluate the practicality and viability of a proposed project. The prime goal of a feasibility study is to determine whether the project is worth pursuing and if it can be successfully implemented within the defined constraints. Each structure has to be thought of in the developing of the project, as it has to serve the end user in friendly manner. Three considerations involved in feasibility are

- Technical Feasibility
- Operational Feasibility
- Economic Feasibility

2.2.1 TECHNICAL FEASIBILITY

Technical feasibility refers to evaluating whether the required technology, tools and resources are available to successfully develop and implement the proposed system. For this project the Student Registration Tracking System is developed using the Salesforce platform, which provides built-in features such as Experience Cloud, Flows, Approval Processes, Reports and Dashboards. These technologies support the development of a web-based system that can manage course approvals, student registrations and monitoring processes efficiently.

The Salesforce Developer Edition provides a cloud-based environment that allows easy configuration and development without requiring complex hardware or software installation. The platform also supports customization through objects, automation through flows and data analysis through reports and dashboards. These features make it technically feasible to implement the entire system within Salesforce.

Since the system is cloud-based it can be accessed through a web browser and does not require high-end infrastructure or specialized technical equipment. The availability of Salesforce documentation, development tools and user-friendly configuration features ensures that the system can be developed and maintained effectively making the project technically feasible.

2.2.2 OPERATIONAL FEASIBILITY

Operational feasibility refers to how well the proposed system will work in the real working environment and whether it will be accepted and used by the users. The Student Registration Tracking System is designed to simplify the process of course management and student registration for the placement department. The system provides an easy-to-use interface where administrators can create courses the placement head can approve them and students can register through the website without any complex procedures.

Since the platform is developed using Salesforce Experience Cloud users can access the system through a web browser making it convenient to operate. The automated processes such as course approval and email notifications reduce manual work and help the placement department manage training activities more efficiently. Therefore the system is operationally feasible and suitable for practical use.

2.2.3 ECONOMIC FEASIBILITY

Economic feasibility refers to analyzing whether the proposed system is cost effective and beneficial for the organization. The Student Registration Tracking System is developed using Salesforce Developer Edition which is available for free for learning and development purposes. Since the system is cloud-based there is no need to purchase additional hardware or install complex software which reduces the overall implementation cost.

The system also helps reduce manual work involved in managing course registrations and student data. Automation of processes such as course approval, email notifications and registration tracking saves time and improves efficiency for the placement department. Therefore the proposed system is economically feasible as it provides better management and productivity with minimal cost.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed Student Registration Tracking System. It describes how the system should operate, including course creation, approval, and student registration. The SRS also identifies the users involved in the system such as administrator, placement head, placement faculty and students. The document acts as a guideline for system design, development and testing to ensure the system meets project requirements.

2.3.1 HARDWARE REQUIREMENTS

The following hardware requirements are necessary for the development and execution of the proposed system:

- **Device Type:** Laptop / Desktop Computer
- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher
- **Processor Speed:** 2.0 GHz or higher
- **Installed RAM:** 4 GB minimum (8 GB recommended)
- **System Type:** 64-bit Operating System, x64-based Processor
- **Storage:** Minimum 256 GB Hard Disk / SSD
- **Display Resolution:** 1366 × 768 pixels or higher

2.3.2 SOFTWARE REQUIREMENTS

The following software requirements are necessary for the development and execution of the proposed system:

- **Operating System:** Windows 10 / Windows 11
- **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
- **Platform:** Salesforce Developer Edition
- **Development Environment:** Salesforce Setup and Experience Builder
- **Technologies Used:** Experience Cloud, Salesforce Flows, Approval Process
- **Reporting Tools:** Salesforce Reports and Dashboards
- **Programming Support:** Salesforce Lightning Platform

2.4 SOFTWARE DESCRIPTION

2.4.1 FRONT END

The front end of the Student Registration Tracking System is developed using Salesforce Experience Cloud and Experience Builder which provides an interactive and user-friendly interface for users to access the system through a web browser. The Experience Cloud website includes several pages such as Home, Courses, Registration, Student, Reports and Placement Head allowing users to easily navigate through different sections of the application. The navigation menu helps users move between pages smoothly while the interface design ensures that information such as course details and registration options are clearly displayed. This front-end environment acts as the main platform where students and placement staff interact with the system.

The front end mainly allows students to view available courses and register for them through the website using the registration form. Students can access the registration page through the email link sent by the system and submit their details without logging into Salesforce. Placement Head and Placement Faculty can access specific pages such as reports and dashboards after login to monitor course registrations and student participation. Using Experience Builder components, forms and page layouts the interface is designed to be simple, organized and easy to use enabling smooth interaction between users and the Salesforce system.

2.4.2 BACK END

The back end of the Student Registration Tracking System is implemented using the Salesforce platform which manages the system logic, automation and data processing. Salesforce tools such as Flows, Approval Processes and system configurations are used to control how the system functions behind the user interface. The back end is responsible for handling operations such as course creation, course approval, sending registration emails to students and storing registration details in the database.

Automation plays an important role in the back end of the system. Salesforce Flows are used to automatically send registration emails to students once a course is approved by the Placement Head.. The back end also manages the interaction between different objects such as Students, Courses and Course Registrations ensuring that all data is stored and processed correctly within the Salesforce system**.**

2.4.3 DATABASE

The database of the Student Registration Tracking System is managed using Salesforce Objects which store and organize all the data required for the system. In Salesforce objects function similar to database tables where records are stored in a structured format. The system uses custom objects to maintain important information related to students, courses and course registrations.

The main database objects used in this project are Students, Courses and Course Registrations. The Students object stores student details such as name, email and phone number while the Courses object stores course information including course name, description and approval status. These objects are connected through relationships allowing the system to efficiently manage and track student registrations within the Salesforce platform.

SUMMARY

This chapter explained the system analysis and requirements. It discussed the identification of the need for an automated system to manage course registrations efficiently. The feasibility analysis including technical, operational, and economic feasibility was also described. The chapter further explained the software requirement specifications along with hardware and software requirements.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

A module description provides a concise overview of a module's purpose and functionality within a system. It outlines the key features, dependencies and potential use cases to help users understand its role. In software development a module description explains how the module interacts with other components and any required configurations. These modules work together to ensure that the system operates smoothly and that all activities related to course registration are properly managed.This project consists of the following modules:

- User Management Module
- Course Management Module
- Course Approval Module
- Student Registration Module
- Email Notification Module
- Reports and Dashboard Module

3.1.1 User Management Module

The User Management Module is responsible for managing different users involved in the Student Registration Tracking System. The system includes various users such as the System Administrator, Placement Head, Placement Faculty, and Students. Each user is assigned specific roles and permissions based on their responsibilities in the system. This module ensures that only authorized users can perform specific actions such as course approval, registration monitoring and report viewing. The administrator manages user accounts and controls access to different features of the system. By managing user roles and access permissions the module helps maintain security and proper functioning of the system.

3.1.2 Course Management Module

The Course Management Module is responsible for creating and managing course details in the Student Registration Tracking System. This module is mainly handled by the System Administrator who enters information such as course name, description and course status. The administrator can update or manage course details whenever necessary. Once a course is created it is submitted for approval to the Placement Head. The module ensures that all course information is properly maintained in the system. This helps in organizing and managing the available courses efficiently. It also allows the administrator to control which courses are available in the system. The module supports proper record management for all created courses. It helps maintain a structured database of course information for future reference.

3.1.3 Course Approval Module

The Course Approval Module manages the process of approving or rejecting courses before they are made available to students. After the administrator creates a course it is submitted for approval to the Placement Head through the approval process. The Placement Head reviews the course details and decides whether to approve or reject the course. If the course is approved it becomes available for student registration. If it is rejected the course will not be visible to students. Once the course is approved the system can trigger automated actions such as sending registration emails to students. This module ensures that only authorized and verified courses are offered through the system and maintains proper control over course availability.

3.1.4 Student Registration Module

The Student Registration Module allows students to register for available courses through the Experience Cloud website. Students can access the registration page using the link provided in the email notification. They can view the course details and submit their information through the registration form. The system stores the registration details in the Course Registrations object within Salesforce. This module helps in maintaining accurate records of students who have registered for each course. It also allows the placement department to track and manage student participation effectively.

3.1.5 Email Notification Module

The Email Notification Module is responsible for sending automated emails to students regarding course registration. When a course is approved by the Placement Head the system triggers a flow that sends email notifications to all students. The email contains important information such as the course name and the registration link. This allows students to easily access the registration page and enroll in the course. The module helps ensure that students receive timely updates about available courses. It also reduces manual work for the placement department by automating the communication process. The module supports sending notifications to multiple students at the same time. It helps improve communication between the placement department and students.

3.1.6 Reports and Dashboard Module

The Reports and Dashboard Module helps in monitoring and analyzing the course registration data in the system. It allows the Placement Head and Placement Faculty to view reports related to courses and student registrations. Reports such as course registration report, student registration report and course-wise student count can be generated. The dashboard presents this information in a visual format using charts and graphs. This makes it easier for users to understand the registration statistics. The module helps the placement department track student participation and evaluate course performance. It also supports better decision making by providing clear insights into the system data. Additionally, reports can be subscribed so that the Placement Head receives updates through email. The module also helps in maintaining proper records of course registrations for future reference.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that shows the flow of data through a system. It illustrates how data moves from one entity to another, how it is processed, and how it is stored. DFDs help in understanding the overall system functionality and interactions between external entities and internal processes. It also helps in identifying the inputs and outputs of different processes within the system. DFDs provide a clear visual representation that makes complex systems easier to understand.

3.2.1 DFD Level 0

A Data Flow Diagram (DFD) Level 0 also known as a Context Diagram represents the overall view of a system. It shows the system as a single process and illustrates how external entities interact with it. The diagram focuses on the flow of data between the system and external users without showing detailed internal processes. DFD Level 0 helps in understanding the basic functionality of the system and identifies the main inputs and outputs involved in the system.

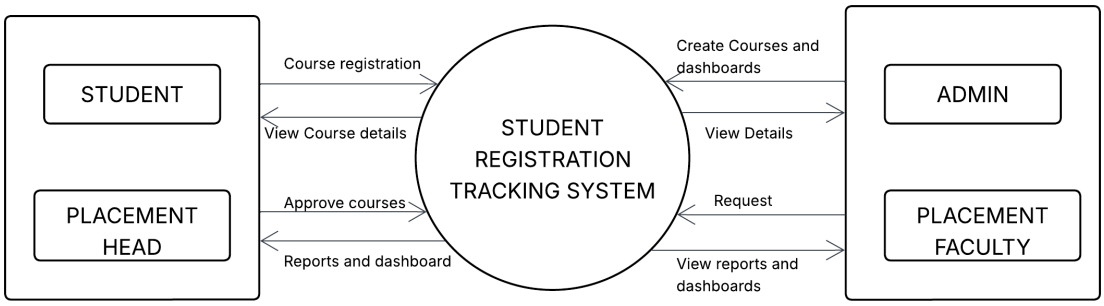


Figure3.1 DFD Level 0

In the Figure 3.1 illustrates the system interacts with four external entities: Student, Placement Head, Admin and Placement Faculty. Students interact with the system by viewing course details and registering for courses. The Placement Head approves courses and views reports and dashboards related to course registrations. The admin manages the system by creating courses and dashboards and viewing system details. The Placement Faculty can send requests and view reports and dashboards to monitor student registrations.

3.2.2 DFD Level 1

A DFD Level 1 (Data Flow Diagram Level 1) is a detailed diagram that expands the processes shown in the Level 0 DFD by breaking the main process into smaller sub-processes. It shows how data flows between different processes, external entities, and data stores within the system. Level 1 DFD helps to clearly understand how the system functions internally and how data is processed step by step. It also helps in identifying the individual operations performed by the system. The diagram provides a clearer representation of how different components of the system interact with each other. It is mainly used during system design to analyze and understand the detailed workflow of the system.

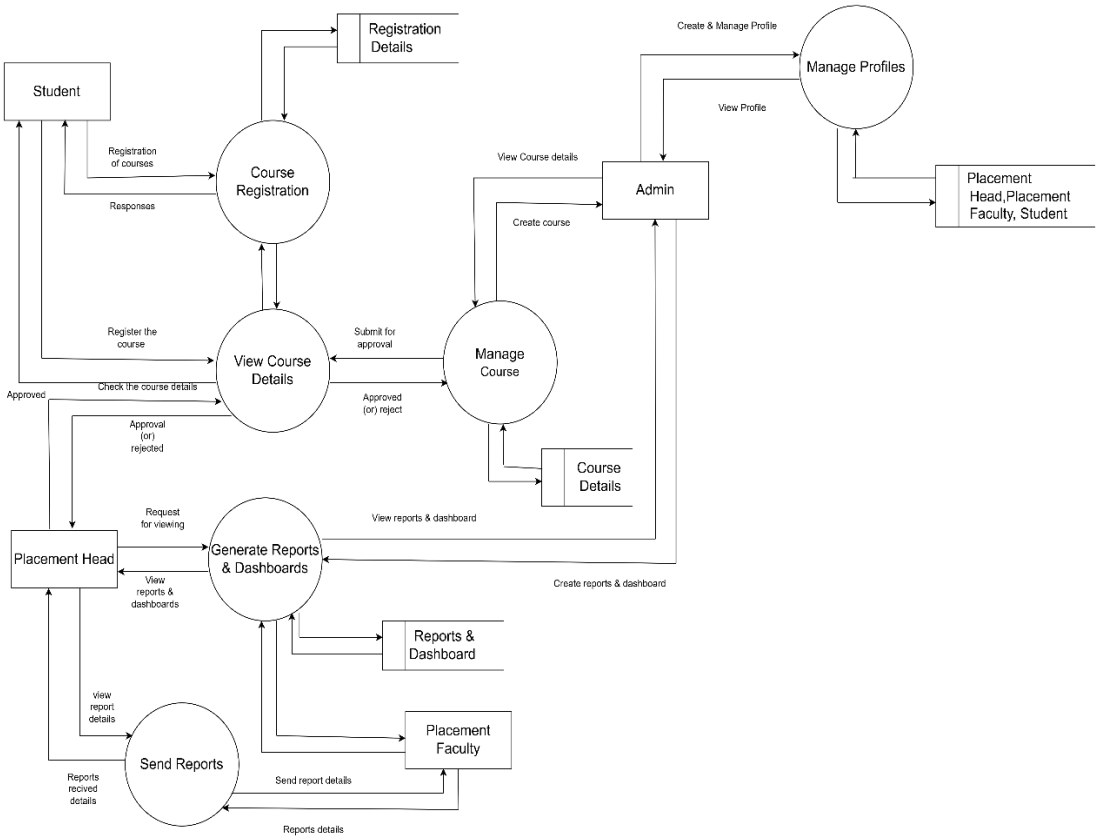


Figure 3.2 DFD level 1

In the Figure 3.2 The Level 1 Data Flow Diagram (DFD) of the Student Registration Tracking System provides a detailed representation of how data flows between users, processes and data stores within the system. The main external entities involved in the system are Student, Admin, Placement Head and Placement Faculty. Each of these users interacts with different system processes to perform specific tasks. The diagram shows how data is processed, stored and transferred between different components of the system.

In the system the Student interacts with processes such as View Course Details and Course Registration. Students can view the available courses and register for them by submitting their details. The registration information is stored in the Registration Details data store. The Admin is responsible for managing the courses through the Manage Course process where course information is created, updated and stored in the Course Details database. The admin also manages user roles and permissions through the Manage Profiles process which maintains profiles such as Placement Head, Placement Faculty and Student.

The Placement Head and Placement Faculty mainly focus on monitoring and analyzing the system activities. The Placement Head generates Reports and Dashboards to track student registrations and course participation. These reports are stored in the Reports & Dashboard data store and can be shared with the Placement Faculty through the Send Reports process. This helps placement staff monitor course performance and student involvement effectively. Overall, the Level 1 DFD explains how the system processes data and supports the interaction between different users and system components.

3.3 USECASE DIAGRAM

A Use Case Diagram is a type of UML (Unified Modelling Language) diagram used to represent the interactions between users (actors) and the system. It shows the different functionalities of the system and how various users interact with those functions. The diagram helps in identifying the roles of different users and the tasks they can perform within the system. It provides a clear and simple overview of system operations and user activities. Use case diagrams are mainly used during the system design phase to understand system requirements and user interactions. They also help developers and stakeholders visualize how the system will operate and how different users will access system features.

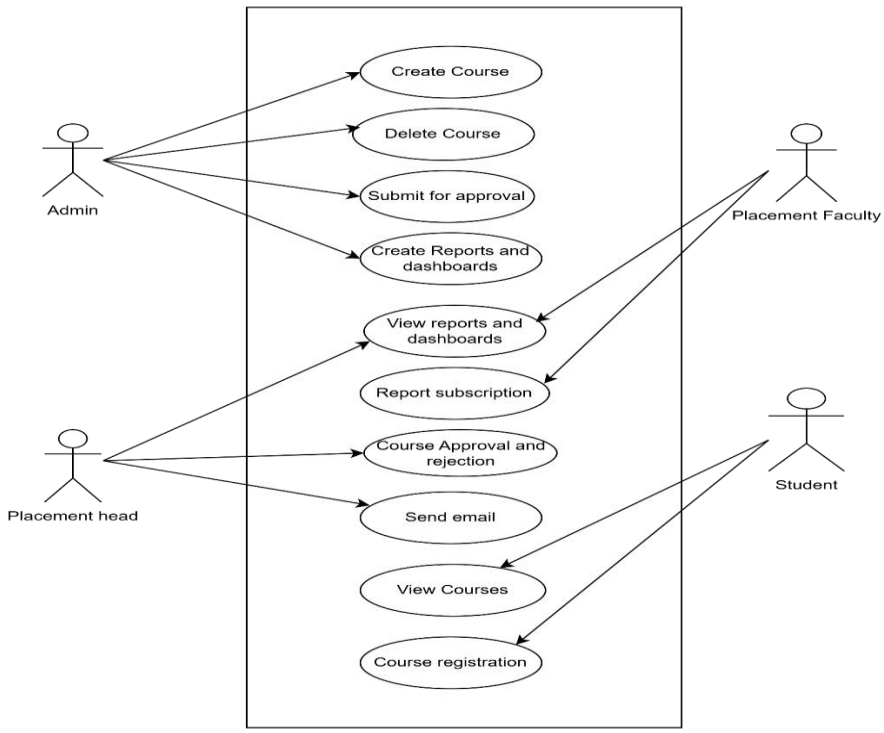


Figure 3.3 use case diagram

In Figure3.3 The Use Case Diagram illustrates the interactions between different users and the Student Registration Tracking System. The main actors involved in the system are Admin, Placement Head, Placement Faculty and Student. The Admin is responsible for managing course-related activities such as creating courses, deleting courses, submitting courses for approval and generating reports and dashboards. These actions help in organizing and maintaining course information within the system. The admin also ensures that the course details are properly stored and available for further approval and monitoring.

The Placement Head reviews the courses submitted by the admin and performs course approval or rejection based on the course requirements. After approving a course the Placement Head can send email notifications to students informing them about the available courses. The Placement Head can also subscribe to reports and dashboards to monitor student registrations and course performance. The Placement Faculty mainly monitors the system by viewing reports and dashboards to track student participation. The Student can view available courses and register for them through the system.

3.4 INPUT DESIGN

Input design refers to the method of collecting and entering data into the system in an organized and accurate manner. In the Student Registration Tracking System, input design focuses on gathering necessary information from users such as administrators and students. The administrator enters course details like course name, description and status while creating a course in the system. Students provide their personal details such as name, email and phone number during the course registration process. The input forms are designed to be simple and easy to use so that users can enter data without difficulty.

Proper validation is applied to ensure that only correct and complete information is stored in the system. This helps in maintaining data accuracy and consistency. Effective input design improves the overall performance and reliability of the system.In addition, the input forms in the system include mandatory fields and validation rules to ensure that important information is not left empty. Fields such as student name, email, phone number and course name must be entered before the data is saved in the system. Dropdown menus and date pickers are also used in certain fields such as department, course type and registration date to reduce errors while entering data.

Login Page

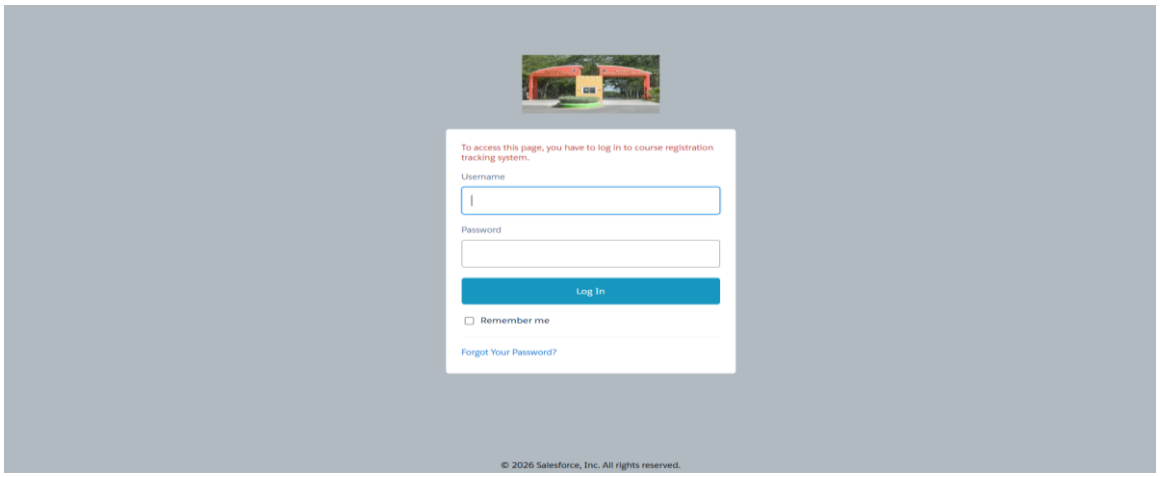


Figure 3.4 Login Page

In the figure 3.4 This page is used to authenticate users before they access the system. It requires users to enter their username and password to log in securely. A message is displayed informing users that they must log in to access the course registration tracking system. The page also includes additional options such as "Remember me", which allows users to stay logged in on the device, and "Forgot Your Password?", which helps users recover their password if they forget it. After entering valid login credentials and clicking the Log In button, authorized users such as Admin, Placement Head and Placement Faculty can access the system and perform tasks like managing courses, approving courses and viewing reports and dashboards.

New Student Registration form

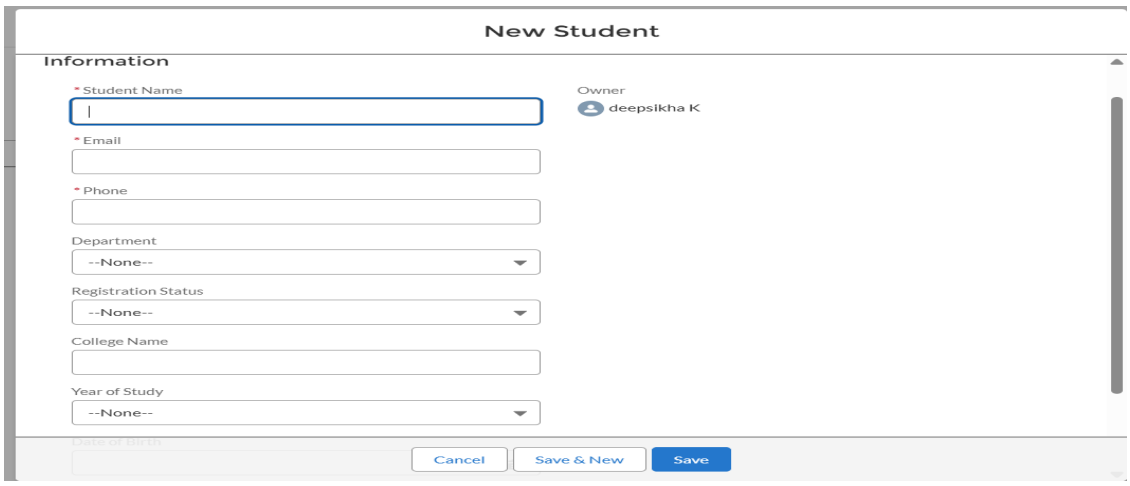


Figure 3.5 New Student Registration form

In Figure 3.5 This form is used to add and store student details in the system database. It contains several fields where the administrator can enter information such as Student Name, Email, Phone number, Department, Registration Status, College Name and Year of Study. Some fields like Student Name, Email and Phone are marked as mandatory, meaning they must be filled before saving the record. The Owner field indicates the user responsible for managing the record. At the bottom of the form, options such as Save, Save & New and Cancel allow the user to either save the student record, save and create another record or cancel the operation. This page helps administrators maintain and manage student information required for course registration within the system.

New Course registration form

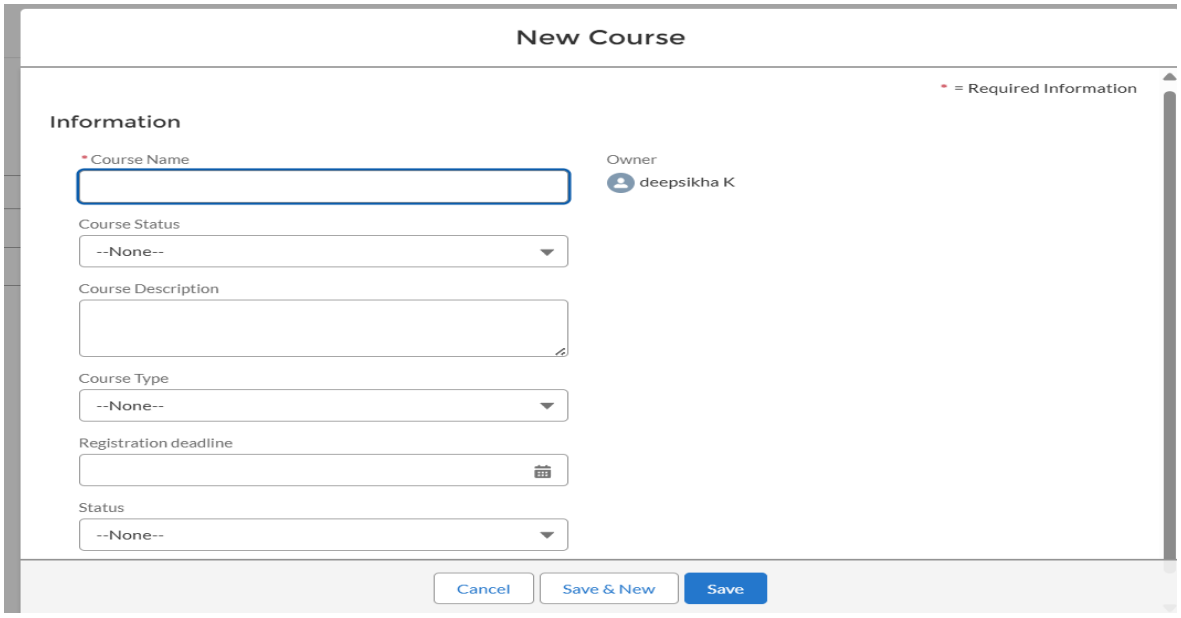


Figure 3.6 New Course registration form

In Figure 3.6 this page allows the administrator to create and store course details in the system by entering information such as Course Name, Course Status, Course Description, Course Type, Registration Deadline and Status. The Course Name field is mandatory which means it must be filled before saving the record. The Owner field indicates the user responsible for managing the course record. At the bottom of the form, options such as Save, Save & New and Cancel allow the user to save the course details, save and create another course or cancel the operation. This page helps administrators manage and organize course information before it is submitted for approval and made available for student registration.

Students Course Registration form

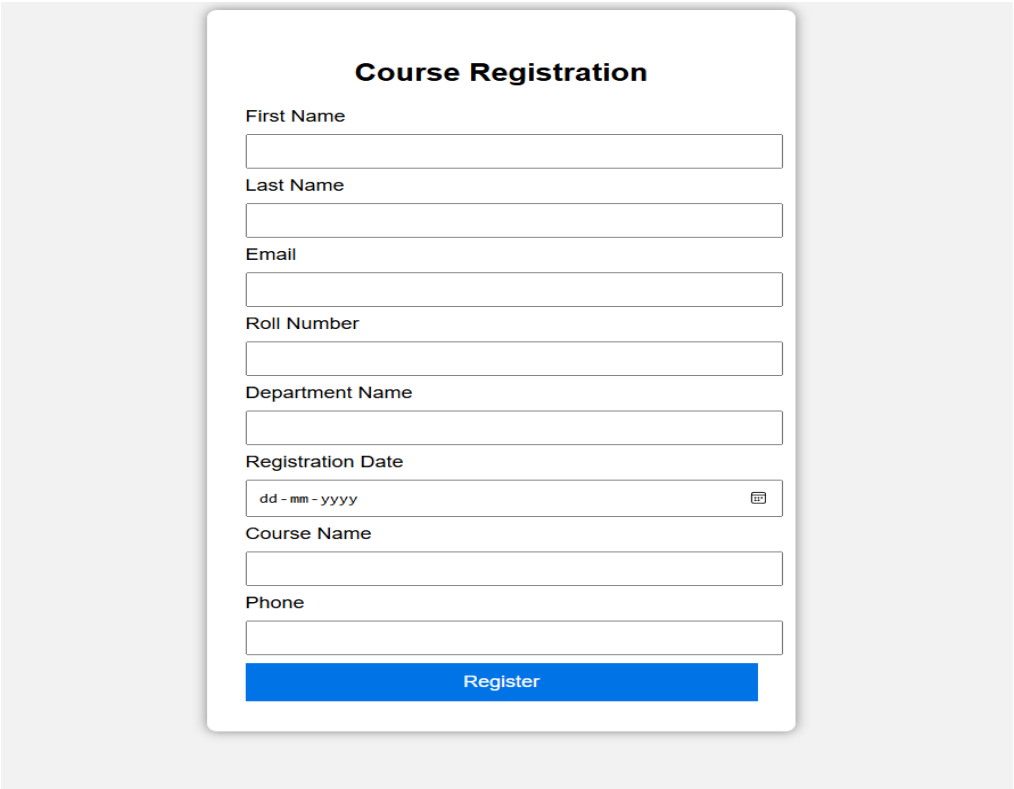


Figure 3.7 Students Course Registration form

In figure 3.7 this form allows students to enter their details to register for a specific course. It includes fields such as First Name, Last Name, Email, Roll Number, Department Name, Registration Date, Course Name and Phone Number. Students must fill in these details correctly to complete the registration process. After entering the required information the student clicks the Register button to submit the form. The submitted details are then stored in the system database allowing the system to track student registrations for different courses.

3.5 OUTPUT DESIGN

Output design refers to the process of presenting the information generated by the system in a clear and organized format so that users can easily understand it. In the Student Registration Tracking System, the output includes details such as course information, student registration records, course approval status and reports or dashboards. These outputs help different users such as administrators, placement heads, faculty members and students to access the required information efficiently.

Home Page

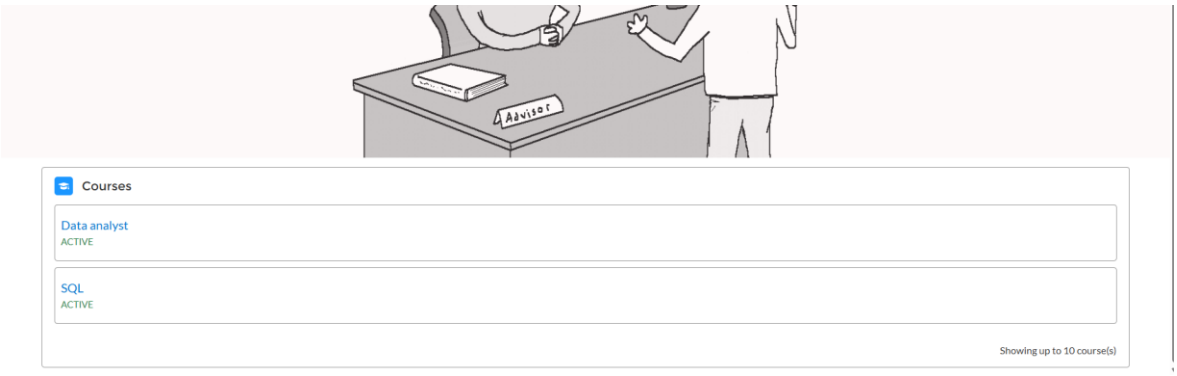


Figure 3.8 Home Page

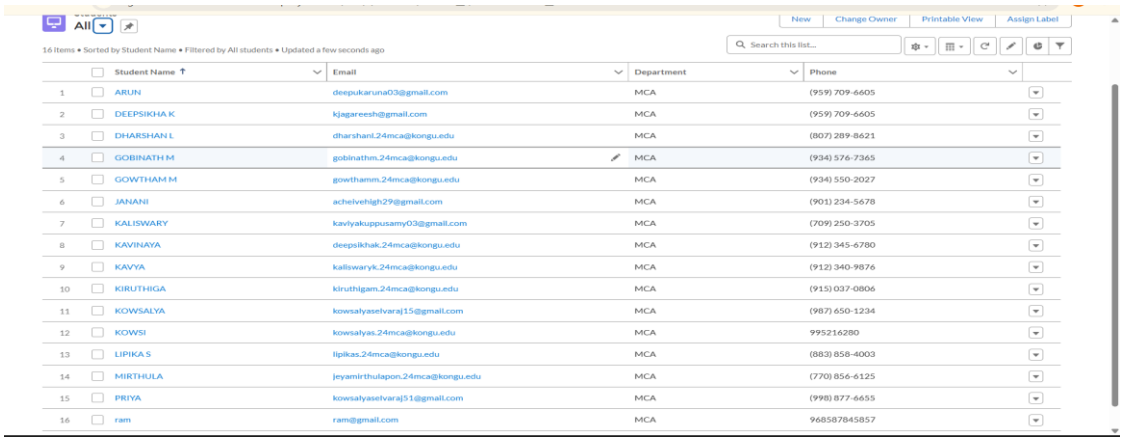
In Figure 3.8 shows the Courses page in the Student Registration Tracking System portal, where the available courses are displayed for users to view. Each course is shown in a card format with the course name and its status such as Data Analyst and SQL both marked as Active. This page allows students or users to easily see the courses that are currently available for registration. The system displays only the active courses so that students can select and register for them.

Courses Approved list

Figure 3.9 Courses Approved List

In figure 3.9his page allows administrators to view, manage and organize course records stored in the system. Each row represents a course such as Data Analyst, Java Programming, Python Programming, and SQL and the courses are sorted by course name. The page also provides options such as New, Change Owner, Printable View and Assign Label which allow users to create new courses, modify course ownership, print the course list or categorize records.

Students list



All

New | Change Owner | Printable View | Assign Label

16 Items • Sorted by Student Name • Filtered by All students • Updated a few seconds ago

Search this list...

	Student Name	Email	Department	Phone
1	ARUN	deepukaruna03@gmail.com	MCA	(959) 709-6605
2	DEEPSIKHA K	kjagareesh@gmail.com	MCA	(959) 709-6605
3	DHARSHAN L	dharshanl.24mca@kongu.edu	MCA	(807) 289-8621
4	GOBINATH M	gobinathm.24mca@kongu.edu	MCA	(934) 576-7365
5	GOWTHAM M	gowthamm.24mca@kongu.edu	MCA	(934) 550-2027
6	JANANI	achelvehigh29@gmail.com	MCA	(901) 234-5678
7	KALISWARY	kaviyakuppusamy03@gmail.com	MCA	(709) 250-3705
8	KAVINAYA	deepsikhak.24mca@kongu.edu	MCA	(912) 345-6780
9	KAVYA	kaliswaryk.24mca@kongu.edu	MCA	(912) 340-9876
10	KIRUTHIGA	kiruthigam.24mca@kongu.edu	MCA	(915) 037-0806
11	KOWSALYA	kowsalyaselvaraj15@gmail.com	MCA	(987) 650-1234
12	KOWSI	kowsalyas.24mca@kongu.edu	MCA	995216280
13	LIPIKA S	lipikas.24mca@kongu.edu	MCA	(883) 858-4003
14	MIRTHULA	jeyamirthulapon.24mca@kongu.edu	MCA	(770) 856-6125
15	PRIYA	kowsalyaselvaraj51@gmail.com	MCA	(998) 877-6655
16	ram	ram@gmail.com	MCA	968587845857

Figure 3.10 Students list

The Figure 3.10 shows the Students list page in the Salesforce Student Registration Tracking System where all student records stored in the system are displayed in a tabular format. Each row represents a student and includes details such as Student Name, Email, Department and Phone number. The records are sorted by Student Name and allow administrators to easily view and manage student information. This page also provides options for searching, filtering, editing and updating student records when necessary.

Student Report Dashboard

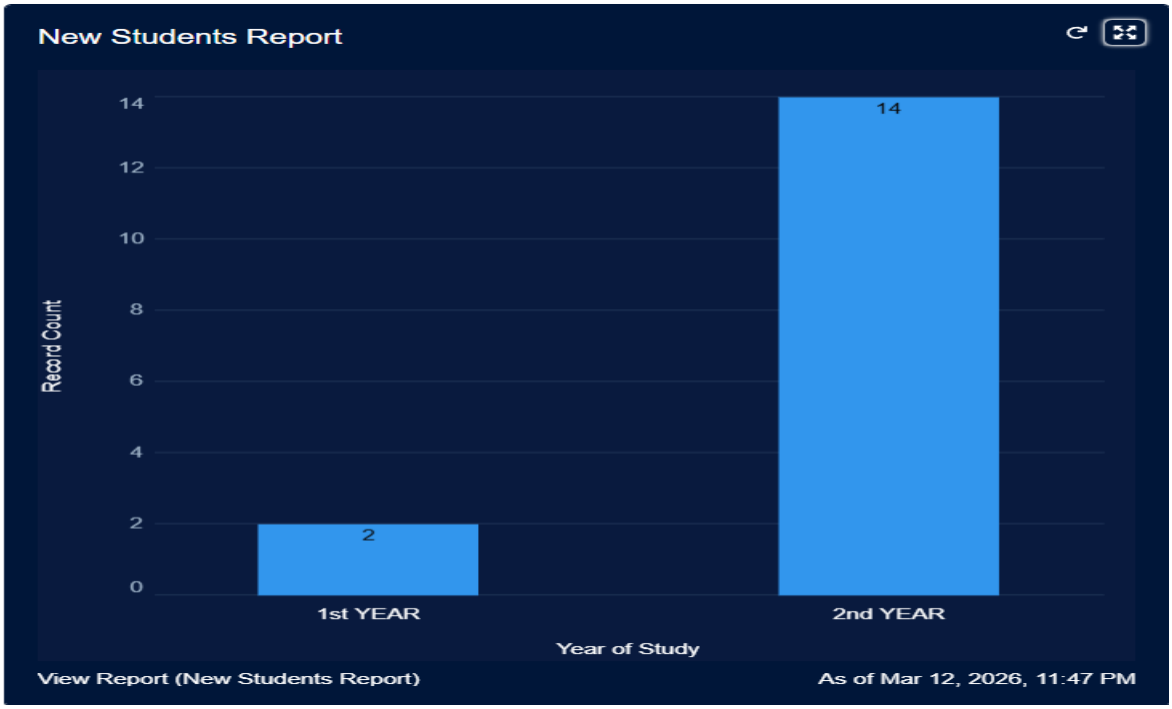


Figure 3.11 Students Report Dashboard

The Figure 3.11 The New Students Report dashboard displays the number of students in the system based on their year of study. The data is represented using a bar chart where the horizontal axis shows the Year of Study and the vertical axis shows the Record Count of students. According to the chart 2 students belong to the 1st year while 14 students belong to the 2nd year indicating that a larger number of registered students are from the second year. This dashboard helps administrators and placement staff easily analyze the distribution of students by their academic year and monitor student participation in the system.

New Leads Dashboard

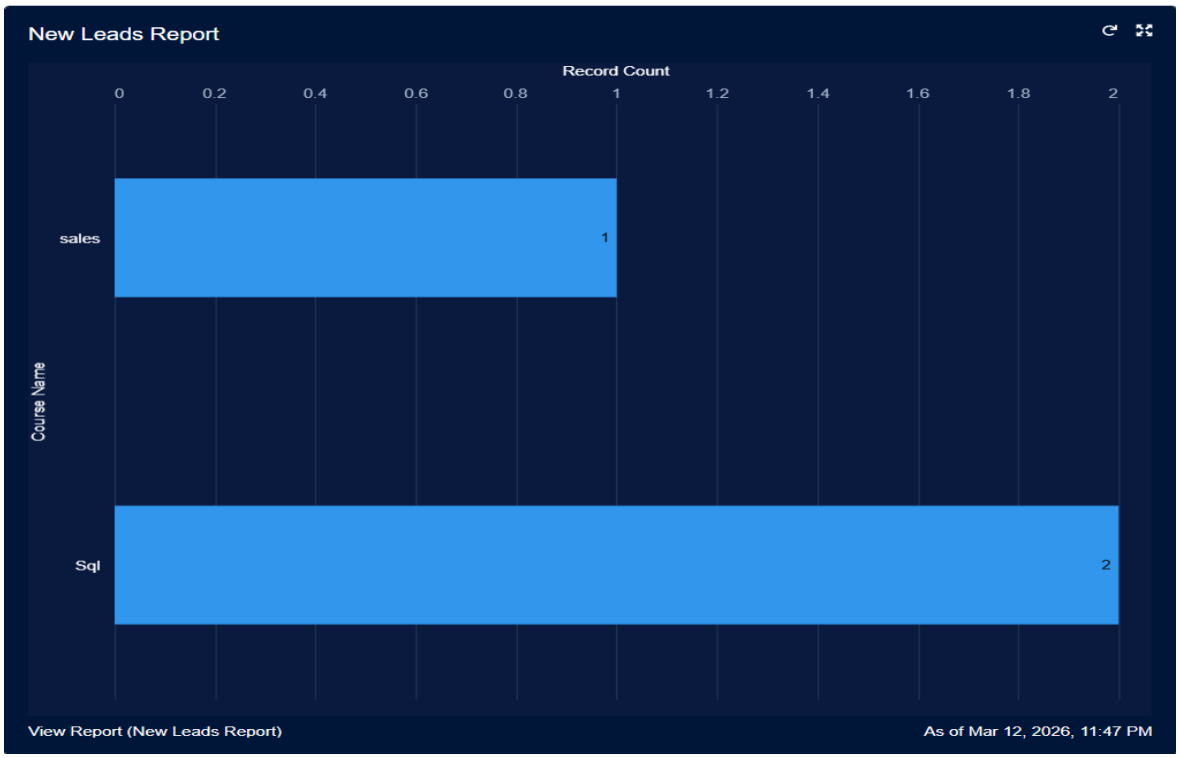
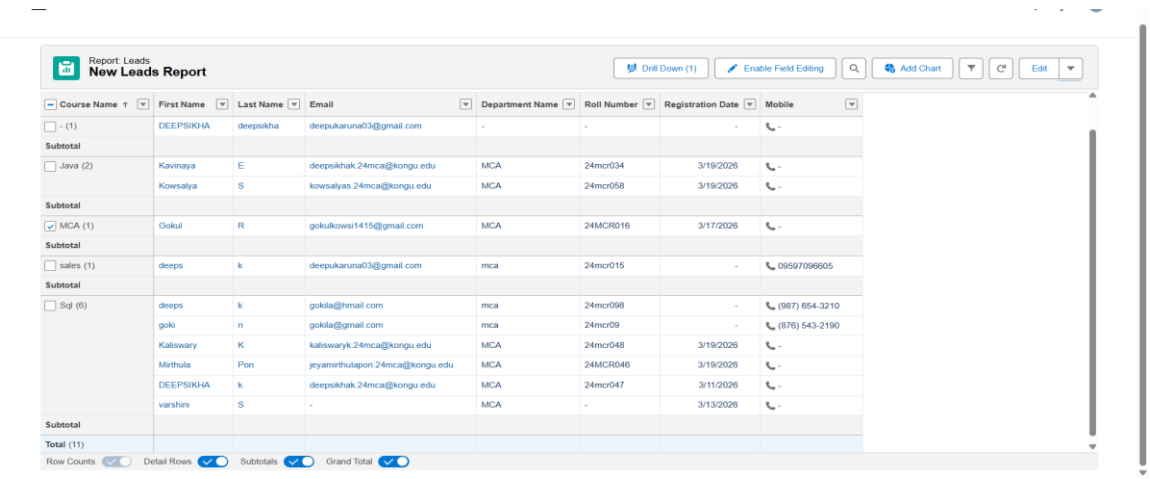


Figure 3.12 New Leads Dashboard

In figure 3.12 The New Leads Dashboard displays the number of records associated with different courses in the Student Registration Tracking System. The report is presented using a bar chart where the horizontal axis represents the record count and the vertical axis represents the course names. In this report the course SQL has 2 records indicating a higher number of entries while the course Sales has 1 record. This visual representation helps administrators easily compare the number of registrations or records associated with each course and understand which courses have more participation.

Leads report



Report: Leads
New Leads Report

Drill Down (1) | Enable Field Editing | Add Chart | Edit

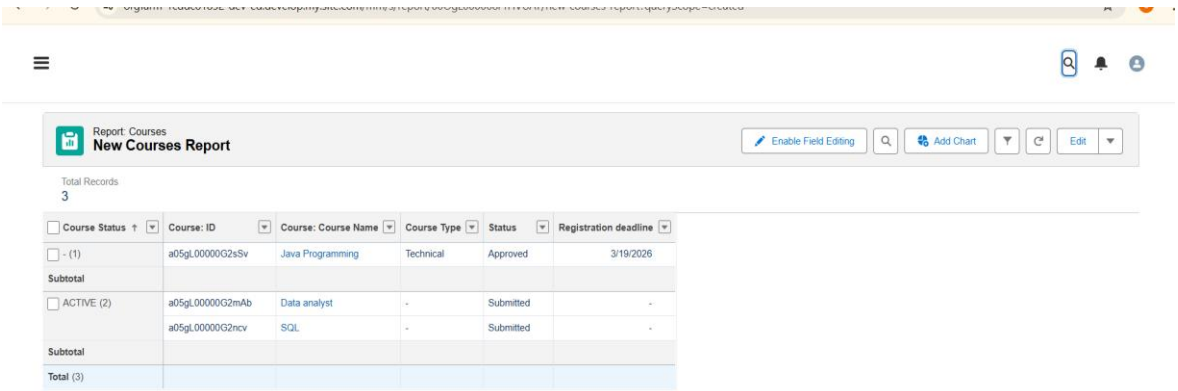
Course Name	First Name	Last Name	Email	Department Name	Roll Number	Registration Date	Mobile
- (1)	DEEPSIKHA	deepsikha	deepukaruna03@gmail.com	-	-	-	-
Subtotal							
Java (2)	Kavinaya	E	deepsikhak.24mca@kongu.edu	MCA	24mcr034	3/19/2026	-
	Kowsalya	S	kowsalyas.24mca@kongu.edu	MCA	24mcr058	3/19/2026	-
Subtotal							
MCA (1)	Gokul	R	gokulkowsi1415@gmail.com	MCA	24MCR016	3/17/2026	-
Subtotal							
sales (1)	deeps	k	deepukaruna03@gmail.com	mca	24mcr015	-	09597096605
Subtotal							
Sql (6)	deeps	k	gokila@hmail.com	mca	24mcr098	-	(987) 654-3210
	goki	n	gokila@gmail.com	mca	24mcr09	-	(876) 543-2190
	Kaliswary	K	kaliswaryk.24mca@kongu.edu	MCA	24mcr048	3/19/2026	-
	Mirthula	Pon	jeyamirthulapon.24mca@kongu.edu	MCA	24MCR046	3/19/2026	-
	DEEPSIKHA	k	deepsikhak.24mca@kongu.edu	MCA	24mcr047	3/11/2026	-
	varshini	S	-	MCA	-	3/13/2026	-
Subtotal							
Total (11)							

Row Counts | Detail Rows | Subtotals | Grand Total

Figure 3.13 Leads Report

In figure 3.13 The image shows the New Leads Report in the Student Registration Tracking System, which displays detailed information about students who have registered for different courses. The report is presented in a tabular format and includes fields such as Course Name, First Name, Last Name, Email, Department Name, Roll Number, Registration Date and Mobile number. The records are grouped based on the Course Name, allowing users to easily see how many students have registered for each course. Subtotals are shown for each course category, and a grand total of 11 records is displayed at the bottom of the report.

Course report



Report: Courses
New Courses Report

Enable Field Editing | Add Chart | Edit

Total Records
3

Course Status	Course: ID	Course: Course Name	Course Type	Status	Registration deadline
- (1)	a05gL00000G2sSv	Java Programming	Technical	Approved	3/19/2026
Subtotal					
ACTIVE (2)	a05gL00000G2mAb	Data analyst	-	Submitted	-
	a05gL00000G2ncv	SQL	-	Submitted	-
Subtotal					
Total (3)					

Figure 3.14 Course report

The figure 3.14 the report displays fields such as Course Status, Course ID, Course Name, Course Type, Status and Registration Deadline. The records are grouped based on the Course Status, allowing users to easily view the status of each course. In this report there are three total course records, where one course (Java Programming) has been approved and two courses (Data Analyst and SQL) are in the submitted status. Subtotals are displayed for each category and a total of three courses is shown at the bottom. This report helps administrators and placement heads monitor the course approval process and track the status of courses in the system.

3.6 ER DIAGRAM

An Entity Relationship (ER) Diagram is a graphical representation used to illustrate the structure of a database. It shows the entities involved in the system their attributes and the relationships between them. ER diagrams help in understanding how data is organized and how different entities interact with each other within the system. They are commonly used during the system design phase to model the database structure clearly and efficiently.

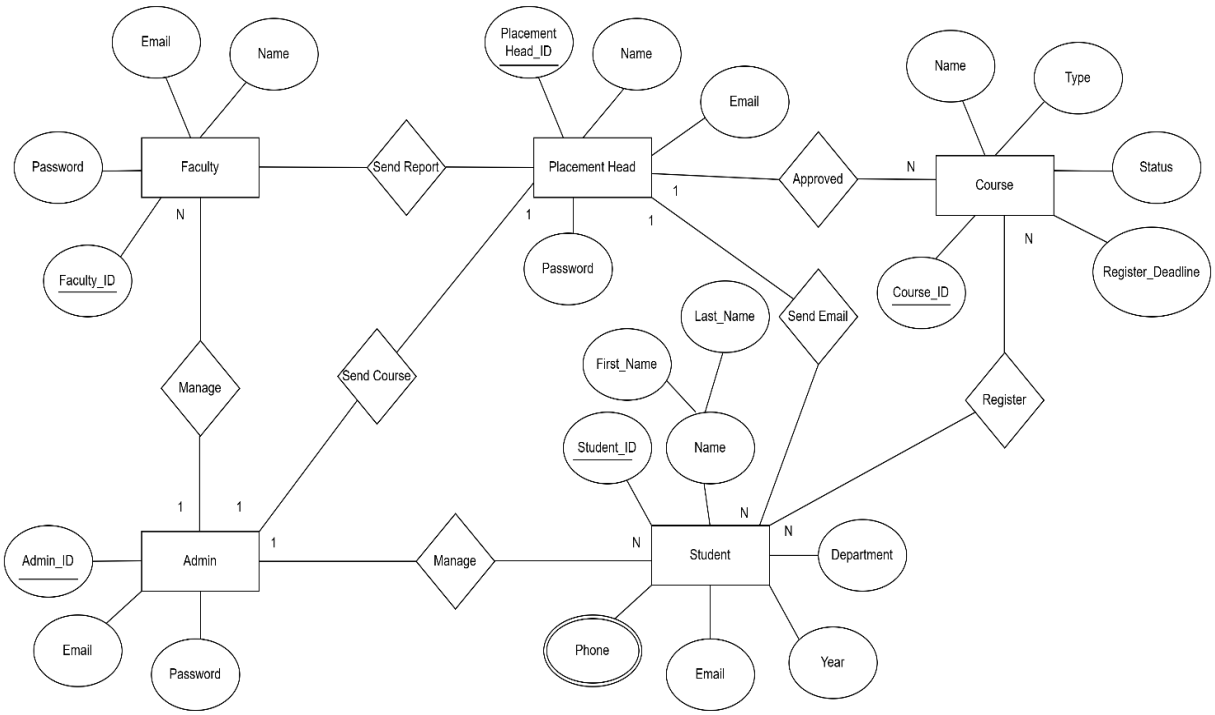


Figure 3.15 ER Diagram

In Figure 3.15 The Entity Relationship (ER) diagram represents the database structure of the Student Registration Tracking System by illustrating the main entities, their attributes and the relationships between them. The primary entities involved in the system include Admin, Faculty, Placement Head, Student and Course. Each entity contains specific attributes that store relevant information. For instance, the Student entity includes attributes such as Student_ID, First_Name, Last_Name, Name, Email, Phone, Department and Year. Similarly, the Course entity contains attributes such as Course_ID, Name, Type, Status and Register_Deadline. The Admin, Faculty, and Placement Head entities also include attributes like ID, Email, Name and Password which help in identifying users and controlling system access.

The Admin is responsible for managing the system and sending course details while the Faculty manages course-related activities and sends reports to the Placement Head. The Placement Head reviews and approves the courses before they are made available to students and also sends email notifications regarding course availability. The Student entity is connected to the Course entity through the Register relationship which represents the course registration process.

3.7 Database Design

Database design refers to the process of organizing and structuring the data used in the system. In the Student Registration Tracking System, the database is designed to store and manage information related to students, courses, registrations and system users. The database uses different entities such as Student, Course, Admin, Placement Head and Placement Faculty, each containing specific attributes that store important details required for the system. These attributes include information such as student details, course information, approval status and registration records

The database is structured in a way that establishes relationships between these entities to ensure proper data management and retrieval. For example the Student entity is connected to the Course entity through the registration process, which records the courses selected by students. Similarly, the Admin manages course details and the Placement Head approves or rejects the courses before they become available for student registration. The Placement Faculty can view reports and dashboards generated from the stored data. Proper database design also helps in efficient data retrieval and updating of records whenever necessary.

3.8 TABLE DESIGN

Table 3.1 Student Table

FIELD NAME	DATA TYPE	DESCRIPTION
Student_ID	Primary Key	Unique identifier for each student
Name	Text	Name of the student
Department	Text	Department of the student
Email	Email	Student email address
Date_of_Birth	Date	Date of birth of the student
Year_of_Study	Number	Academic year of the student
College_Name	Text	Name of the college
Phone	Phone	Contact Number of the student

In the Table 3.1, the Student table stores all the important information related to students in the Student Registration Tracking System. Each student is uniquely identified using the Student_ID, which acts as the primary key in the table. The table includes details such as the name, department and email which help identify and communicate with the student. It also contains additional information like date of birth, year of study and college name which represent the academic and personal details of the student.

The phone number field is used to store the contact number of the student for communication purposes. The information stored in this table helps the system maintain proper records of students who participate in course registration. It also contains the datatype of the field name and the description for the field name. This table is needed when a new student record is created.The table helps the system validate and manage student contact information efficiently ensuring that notifications related to course registration, approvals and updates can be communicated to students without errors.

Table 3.2 Course table

FIELD NAME	DATA TYPE	DESCRIPTION
Course_ID	Primary Key	Unique Identifier for each course
Course_Name	Text	Name of the course
Course_Status	Text	Status of the course (Draft, Pending Approval, Approved, Rejected)
Course_Description	Text Area	Detailed description of the course
Course_Type	Text	Type of course
Registration_Deadline	Date	Last date for students to register for the course
Status	Text	Indicates whether the course is active or inactive

In Table 3.2, each course is uniquely identified using the Course_ID, which acts as the primary key in the table. The table includes details such as the course name, course description and course type which help describe the course and its mode of delivery. It also contains the registration deadline which specifies the last date for students to register for the course. The course status and overall status fields indicate whether the course is submitted, approved, rejected or currently active in the system.

This table helps the system maintain and manage all course-related information in an organized manner. The details stored in this table allow administrators to create and update course information, while the placement head can review and approve the courses before they are made available to students. The stored course information is also used to display available courses to students and support the course registration process within the system. It also helps administrators track the progress of courses from submission to approval. This ensures that only approved courses are visible to students for registration. The table helps maintain data consistency and ensures that course details such as course name, duration, properly stored for future reference and reporting purposes.

Table 3.3 Registration Table

FIELD NAME	DATA TYPE	DESCRIPTION
Registration_ID	Primary Key	Unique identifier for each registration
Student Name	Text	Name of the student who registers for the course
Course Name	Text	Name of the Course selected by the student
Registration Date	Date	Date on which the student registers for the course
Department Name	Text	Department of the student
Email_ID	Email	Email address of the student

Each registration is uniquely identified using the Registration_ID, which acts as the primary key in the table. It also records the registration date, which shows when the student completed the registration process. Additional details such as department name and email ID are stored to identify the student and enable communication if needed.

Table 3.4 Admin Table

FIELD NAME	DATA TYPE	DESCRIPTION
Admin_ID	Primary Key	Unique identifier used to distinguish administrator in the system
Name	Text	Email
Email	Email	Used for communication and login
Password	Text	Password used by the administrator to securely access the system

In Table 3.4 the Admin_ID uniquely identifies each administrator in the system. The Name field stores the administrator's name,while the Email field is used for communication and system login purposes. The Password field is used to authenticate the administrator and provide secure access to system functionalities such as creating courses, managing course details and generating reports and dashboards.

Table 3.5 Placement Head Table

FIELD NAME	DATA TYPE	DESCRIPTION
Head_ID	Primary Key	Unique identifier used to distinguish each placement head in the system
Name	Text	Name of the placement Head
Email	Email	Email address used for login and Communication
Password	Text	Used to securely access the system

In Table 3.5, the Head_ID uniquely identifies each placement head in the system. The Name field stores the name of the placement head while the Email field is used for login and communication purposes. The Password field is used to authenticate the placement head and provide secure access to system functions such as approving or rejecting courses, sending email notifications to students and viewing reports and dashboards. These fields ensure that only authorized placement heads can access the system and perform course approval activities.

This table helps the system maintain the details of the placement head responsible for managing course approvals and monitoring system activities. The information stored in this table allows the placement head to review the courses submitted by the administrator and take appropriate action. It also supports communication with students by enabling the placement head to send course registration notifications.

Table 3.6 Placement Faculty Table

FIELD NAME	DATA TYPE	DESCRIPTION
Faculty_ID	Primary Key	Unique identifier used to distinguish each placement faculty member in the system
Name	Text	Name of the placement faculty member
Email	Email	Used for login and communication
Password	Text	Used to securely access the system

In the table 3.6 the Faculty_ID uniquely identifies each faculty member in the system. The Name field stores the name of the placement faculty while the Email field is used for login and communication purposes. The Password field is used to authenticate the faculty member and provide secure access to the system. Placement faculty can view reports and dashboards to monitor student registrations and course participation. The information stored in this table helps in maintaining accurate records of faculty members involved in monitoring the system.This table supports the system by allowing placement faculty to track and analyze course registration activities. Faculty members can use the reports and dashboards to understand student participation in different courses

SUMMARY

This chapter explains the system design of the Student Registration Tracking System. It describes the different modules of the system such as user management, course management, course approval, student registration and reporting. The chapter also includes design diagrams like DFD, ER Diagram and Use Case Diagram to represent the system structure. It explains the input design, output design and database design used in the system. The table design for entities is presented to show how data is stored and managed.

CHAPTER-4

IMPLEMENTATION

4.1 CODE DESCRIPTION

The Student Registration Tracking System is implemented using the Salesforce platform, which mainly uses configuration and automation tools instead of traditional programming code. The system functionality is developed using Salesforce features such as Flows, Approval Processes, Custom Objects and Experience Cloud. These tools help in implementing the system logic required for course management, course approval, student registration and reporting.

In this project Salesforce Flows are used to automate processes such as sending registration emails to students once a course is approved. The Approval Process manages the workflow where the Placement Head can approve or reject the courses created by the administrator. Custom objects like Students, Courses and Course Registrations are used to store and manage the data within the system. Experience Cloud pages provide the interface through which students can view courses and complete their registration.

The system also uses Salesforce Reports and Dashboards to display information about course registrations and student participation. These reports help placement staff monitor the system activity and analyze registration data. The automation and configuration features of Salesforce make it possible to build the system efficiently without complex coding.

4.2 STANDARDIZATION OF THE CODING

Coding standards are strictly followed throughout the development of the Student Registration Tracking System to maintain uniformity, readability and scalability of the system. Since the project is developed using the Salesforce platform, standard configuration practices are applied while creating custom objects, fields, flows and approval processes. Proper organization of fields, page layouts and automation components ensures that the system remains structured and easy to understand.

Following standardized development practices helps keep the system clean, reliable and easy to maintain in the future. Salesforce Flows are designed in a clear and modular structure to automate tasks such as sending email notifications and managing course approval workflows. Validation rules and appropriate field types are used to ensure data accuracy and prevent incorrect entries. Reports and dashboards are also created using proper naming conventions to support easy monitoring of course registrations. This standardized approach improves system reliability, supports future scalability and ensures smooth functioning of the Student Registration Tracking System.

4.3 ERROR HANDLING

Error handling is an important part of the Student Registration Tracking System to ensure that the system operates smoothly and prevents incorrect data from being stored. The system includes validation checks and proper field constraints to ensure that users enter correct and complete information during course creation and student registration. Required fields such as student name, email and course details must be filled before submitting the form which helps reduce data entry errors.

Salesforce validation rules and field data types are used to prevent invalid or incomplete data from being saved in the system. For example, email fields accept only valid email formats and mandatory fields must be completed before submission. In addition Salesforce flows and approval processes are configured to ensure that the system follows the correct workflow when processing course approvals and sending email notifications. These error handling mechanisms help maintain data accuracy and improve the reliability of the system.

SUMMARY

This chapter explained the implementation aspects of the Student Registration Tracking System developed using the Salesforce platform. It described the system configuration, code description and the standards followed during development. The chapter also discussed how standardized practices were used to maintain consistency and organization in the system components. Error handling mechanisms used to maintain data accuracy and prevent invalid inputs were also explained. Overall this chapter provides an understanding of how the system was developed and managed using Salesforce tools and automation features.

CHAPTER 5

TESTING AND RESULTS

5.1 Testing

Testing is an important phase in the development of the Student Registration Tracking System to ensure that the system functions correctly and meets the required objectives. It involves verifying that all modules such as course management, course approval, student registration, email notification and reporting are working properly without errors. Testing helps identify issues in the system and ensures that the system performs its intended functions effectively.

In this project testing is performed to check whether users can successfully create courses, approve or reject courses, send registration emails and allow students to register through the Experience Cloud website. It also verifies that the registration data is stored correctly in the Salesforce database and that reports and dashboards display accurate information. Through proper testing the reliability and performance of the system are ensured before it is fully used. The testing techniques applied in this project include:

- Unit Testing
- Integration Testing
- User Acceptance Testing

5.1.1 UNIT TESTING

Unit testing is performed to test individual modules of the system separately. Each module such as Course Management, Course Approval, Student Registration and Email Notification is tested individually to ensure that it performs its intended function correctly. This testing helps identify errors in specific components before integrating them with other modules. It ensures that each module works independently without affecting other parts of the system. Unit testing also helps improve the reliability and quality of the system.

CASE 1

Module: Course Management Module

Test Type: Course Creation Verification

Input: Course Name, Course Description, Course Status

Output: Course Record Created Successfully

TEST

Input: Valid Course Details entered by Administrator

Output: Course record saved in the Courses object

Analysis: The course details are stored correctly in the system and appear in the course list.

Result: Pass

CASE 2

Module: Course Approval Module

Test Type: Approval Process Verification

Input: Course submitted for approval

Output: Placement Head receives approval request

TEST

Input: Placement Head clicks Approve option

Output: Course status changes to Approved

Analysis: The approval process works correctly and the course status updates successfully.

Result: Pass

CASE 3

Module: Student Registration Module

Test Type: Registration Verification

Input: Student Name, Email, Phone Number, Course Selection

Output: Course Registration Record Created

TEST

Input: Valid student registration details

Output: Student registration stored in Course Registrations object

Analysis: Student registration data is saved correctly and linked to the selected course.

Result: Pass

CASE 4

Module: Email Notification Module

Test Type: Email Flow Testing

Input: Course status changed to Approved

TEST

Input: Placement Head approves the course

Output: Registration email sent to all students with course registration link

Analysis: The flow is triggered successfully after course approval and emails are delivered to the students. The email contains the correct course name and registration link.

Result: Pass

CASE 5

Module: Reports and Dashboard Module

Test Type: Report Generation Verification

Input: Course registration data stored in the system

TEST

Input: Placement Head opens the Course Registration Report

Output: Report displays list of students registered for each course

Analysis: The report correctly retrieves data from the Course Registrations object and displays accurate information about student registrations.

Result: Pass

5.1.2 INTEGRATION TESTING

Integration testing is used to verify that different modules of the system work together properly. In this project it ensures that once a course is approved by the Placement Head, the system correctly triggers the email notification flow and allows students to access the registration page. This testing confirms that the interaction between modules functions smoothly. It verifies that data flows correctly between modules such as course management, approval and student registration. Integration testing also ensures that the system correctly stores registration details after students submit the form. It helps identify issues that occur when different modules interact with each other. This testing improves the overall reliability and coordination between all components of the system.

CASE 1

Modules: Course Management Module → Course Approval Module

Test Type: Module Interaction Verification

Input: Administrator creates a new course

TEST

Input: Course submitted for approval

Output: Placement Head receives course approval request

Analysis: The course created by the administrator is successfully sent to the Placement Head for approval.

Result: Pass

CASE 2

Modules: Course Approval Module → Email Notification Module

Test Type: Automation Trigger Verification

Input: Placement Head approves the course

TEST

Input: Course status changed to Approved

Output: Email notification flow is triggered

Analysis: The system correctly triggers the email flow after the course is approved.
Result: Pass

CASE 3
Modules: Email Notification Module → Student Registration Module
Test Type: Link Redirection Verification
Input: Student receives registration email

TEST
Input: Student clicks the registration link in the email
Output: Student registration page opens in Experience Cloud

Analysis: The email link correctly redirects students to the course registration page.
Result: Pass

CASE 4
Modules: Student Registration Module → Database (Course Registrations Object)
Test Type: Data Storage Verification
Input: Student submits registration form

TEST
Input: Valid student details submitted
Output: Registration record created in Course Registrations object

Analysis: The student registration details are stored correctly in the database.
Result: Pass

CASE 5
Modules: Course Registrations Object → Reports and Dashboard Module
Test Type: Data Retrieval Verification
Input: Student registration data available in the system

TEST

Input: Placement Head opens registration report

Output: Report displays course-wise student registration data

Analysis: The reports correctly retrieve and display the stored registration data.

Result: Pass

5.1.3 USER ACCEPTANCE TESTING

User Acceptance Testing is conducted to ensure that the system meets the requirements of the users. In this stage users such as the Placement Head and Placement Faculty test the system to verify that features like course approval, student registration and report viewing work as expected. This testing confirms that the system is ready for actual use. It helps ensure that the system is user-friendly and easy to operate for different users. User Acceptance Testing also verifies that all system functionalities meet the expected requirements. It allows users to provide feedback about the system performance and usability. This testing ensures that the system works correctly in a real-time working environment before final deployment.

CASE 1

User: System Administrator

Module: Course Management Module

Test Type: Course Creation Verification

TEST

Input: Administrator enters course name, description, and status

Output: Course is created successfully in the Courses object

Analysis: The administrator is able to create and manage course details without errors.

Result: Pass

CASE 2

User: Placement Head

Module: Course Approval Module

Test Type: Approval Process Verification

TEST

Input: Placement Head reviews and approves the course

Output: Course status changes to Approved

Analysis: The Placement Head is able to approve or reject courses successfully.

Result: Pass

CASE 3

User: Student

Module: Student Registration Module

Test Type: Registration Verification

TEST

Input: Student enters name, email, phone number and submits registration form

Output: Course registration record created successfully

Analysis: Students are able to register for courses through the Experience Cloud website.

Result: Pass

CASE 4

User: Student

Module: Email Notification Module

Test Type: Email Notification Verification

TEST

Input: Student receives course registration email and clicks the registration link

Output: Registration page opens successfully

Analysis: Students receive email notifications and can access the registration page without issues.

Result: Pass

CASE 5

User: Placement Faculty

Module: Reports and Dashboard Module

Test Type: Report Access Verification

TEST

Input: Placement Faculty opens reports and dashboards

Output: Course registration reports and dashboard charts are displayed

Analysis: Placement Faculty can view and monitor course registration data successfully.

Result: Pass

CASE 6

User: Placement Head

Module: Reports and Dashboard Module

Test Type: Report Subscription Verification

TEST

Input: Placement Head subscribes to the Course Registration Report

Output: Report is sent automatically to the Placement Head through email at the scheduled time

Analysis: The report subscription works correctly and the Placement Head receives periodic updates about course registrations.

Result: Pass

SUMMARY

This chapter explained the testing process carried out for the Student Registration Tracking System. It described different types of testing such as unit testing, integration testing, and user acceptance testing performed to verify the system functionality. Various test cases were conducted to ensure that modules like course management, course approval, student registration, email notification and reporting work correctly.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1CONCLUSION

The Student Registration Tracking System developed using the Salesforce platform provides an efficient and automated solution for managing course registrations within the placement department. The system simplifies the entire process of course creation, approval, and student registration by using Salesforce features such as Experience Cloud, Flows, Approval Processes, Reports and Dashboards. It replaces the traditional manual methods of managing registrations with a centralized digital system which improves accuracy reduces administrative workload and ensures better organization of course information. The system also allows the System Administrator to create and manage courses while the Placement Head reviews and approves them through a structured approval workflow. This structured process ensures that only authorized courses are made available to students and helps maintain proper control over course management activities.

Once a course is approved the system automatically sends registration emails to students, allowing them to access the registration page through the Experience Cloud website. All registration details are stored in Salesforce objects making it easier to maintain and track student participation. The system also provides reports and dashboards that help placement staff monitor course registrations, analyze student activity and evaluate the effectiveness of training programs. These features support better decision making and provide real-time insights into course participation. In addition, the system improves communication between the placement department and students through automated notifications. Overall, the system enhances transparency, improves efficiency and provides a reliable and scalable platform for managing student course registrations effectively.

6.2 FUTURE ENHANCEMENTS

The Student Registration Tracking System can be enhanced in the future by adding more advanced features to improve its functionality and user experience. One possible improvement is the implementation of automated reminder notifications that inform students about upcoming courses and registration deadlines. This feature will help increase student participation and ensure that students do not miss important training opportunities. The system can also include automated status updates for students after successful registration. Another enhancement could be the addition of course search and filtering options to help students easily find relevant courses. The system can also support online course materials and resources for registered students. Notification alerts can be added to inform users about course updates and announcements. Features such as course capacity management can be implemented to control the number of students allowed in a course. The system can also include automated waitlist management when courses are full. Additionally, a personalized student dashboard can be introduced where students can track their registered courses and updates.

Another enhancement could be the integration of advanced analytics and reporting tools to provide deeper insights into course performance and student participation. The system can also be improved by developing mobile-friendly access so that students and placement staff can use the system easily through smartphones and tablets. Additionally, features such as student feedback for courses, trainer management and automated attendance tracking can be included to make the system more efficient and comprehensive for managing training programs. Integration with other institutional systems can also be added in the future to improve data sharing and system functionality. The system can also include role-based notifications for administrators and placement staff. Security features such as advanced access control and data protection mechanisms can be implemented. Multi-language support can be added to make the system accessible to a wider group of users. Cloud storage integration can also be implemented for managing course documents and training materials. These enhancements will further improve the effectiveness, usability and scalability of the system.

APPENDICES

A.SAMPLE CODING

HOME PAGE

CourselistController.cls

```
public enum COURSE_STATUS {
ACTIVE,
INACTIVE
}
public class CourseItemDTO {
@AuraEnabled public Id id;
@AuraEnabled public String name;
@AuraEnabled public String courseDescription;
@AuraEnabled public String courseStatus;
public CourseItemDTO(Course__c c) {
this.id = c.Id;
this.name = c.Name;
this.courseDescription =
(Schema.sObjectType.Course__c.fields.Course_Description__c.isAccessible()) ?
(String)c.get('Course_Description__c') : null;
this.courseStatus=(Schema.sObjectType.Course__c.fields.Course_Status__c.isAccessible
()) ? (String)c.get('Course_Status__c') : null;
}
}
@AuraEnabled(cacheable=true)
public static List<CourseItemDTO> getActiveCourses(Integer maxItems, String
statusValue) {
Integer limitSize = (maxItems == null || maxItems <= 0 || maxItems > 200) ? 10 :
maxItems;
String statusApi = String.isBlank(statusValue) ? COURSE_STATUS.ACTIVE.name() :
statusValue;
if (!Schema.sObjectType.Course__c.isAccessible()
|| !Schema.sObjectType.Course__c.fields.Name.isAccessible()
```

```
|| !Schema.sObjectType.Course__c.fields.Course_Status__c.isAccessible()) {
return new List<CourseItemDTO>();
}
List<Course__c> courses = [
SELECT Id, Name, Course_Description__c, Course_Status__c
FROM Course__c
WHERE Course_Status__c = :statusApi
WITH SECURITY_ENFORCED
ORDER BY Name ASC
LIMIT :limitSize
];
List<CourseItemDTO> results = new List<CourseItemDTO>();
for (Course__c c : courses) {
results.add(new CourseItemDTO(c));
}
return results;
}
public class GetCoursesRequest {
@InvocableVariable(required=false)
public Integer maxItems;
@InvocableVariable(required=false)
public String statusValue;
}
public class GetCoursesResponse {
@InvocableVariable
public List<CourseItemDTO> courses;
}
@InvocableMethod(label='Get Courses (ACTIVE by default)' description='Returns up to
maxItems Course__c filtered by status (API value), default ACTIVE.')
public static List<GetCoursesResponse> getCoursesInvocable(List<GetCoursesRequest>
requests) {
List<GetCoursesResponse> responses = new List<GetCoursesResponse>();
if (requests == null || requests.isEmpty()) {
```

```
GetCoursesResponse r = new GetCoursesResponse();
r.courses = getActiveCourses(10, COURSE_STATUS.ACTIVE.name());
responses.add(r);
return responses;
}
for (GetCoursesRequest req : requests) {
Integer maxItems = (req == null) ? 10 : req.maxItems;
String statusValue = (req == null) ? COURSE_STATUS.ACTIVE.name() :
req.statusValue;
GetCoursesResponse r = new GetCoursesResponse();
r.courses = getActiveCourses(maxItems, statusValue);
responses.add(r);
}
return responses;
}
}
```

CourseList.html

```
<template>
<article class="slds-card">
<div class="slds-card__header slds-grid">
<header class="slds-media slds-media_center slds-has-flexi-truncate">
<div class="slds-media__figure">
<span class="slds-icon_container slds-icon-standard-education" title="Courses">
<svg class="slds-icon slds-icon_small" aria-hidden="true">
<use xlink:href="/_slds/icons/standard-sprite/svg/symbols.svg#education"></use>
</svg>
</span>
</div>
<div class="slds-media__body">
<h2 class="slds-card__header-title">
<span class="slds-text-heading_small">Courses</span>
```

```
</h2>
</div>
</header>
</div>
<div class="slds-card__body slds-card__body_inner">
<template if:true={isLoading}>
<lightning-spinner size="small" alternative-text="Loading courses"></lightning-spinner>
</template>
<template if:true={error}>
<div class="slds-text-color_error slds-m-bottom_small">
{errorMessage}
</div>
</template>
<template if:true={hasCourses}>
<ul class="slds-has-dividers_around-space">
<template for:each={courses} for:item="c">
<li key={c.id} class="slds-item">
<div class="slds-p-vertical_x-small">
<!-- 1st line: Course Name (clickable) -->
<div class="slds-text-heading_small">
<a href="javascript:void(0)" class="slds-text-link" data-id={c.id}
onclick={handleNavigate}>
{c.name}
</a>
</div>
<template if:true={c.courseStatus}>
<div class="slds-m-top_xxx-small slds-text-color_success slds-text-body_small">
{c.courseStatus}
</div>
```

```
</template>
<template if:true={c.courseDescription}>
<div class="slds-m-top_xxx-small slds-text-color_weak">
{c.courseDescription}
</div>
</template>
</div>
</li>
</template>
</ul>
</template>
<template if:false={hasCourses}>
<template if:false={isLoading}>
<p class="slds-text-color_weak">No courses to display.</p>
</template>
</template>
</div>
<footer class="slds-card__footer">
<span>Showing up to {maxItems} course(s)</span>
</footer>
</article>
</template>
```

CourseList.js

```
import { LightningElement, api, track } from 'lwc';
import { NavigationMixin } from 'lightning/navigation';
import getActiveCourses from '@salesforce/apex/CourseListController.getActiveCourses';
export default class CourseList extends NavigationMixin(LightningElement) {
@api maxItems = 10;
@api statusValue = 'ACTIVE';
```

```
@track courses = [];
@track error;
isLoading = false;
get hasCourses() {
return Array.isArray(this.courses) && this.courses.length > 0;
}
get errorMessage() {
if (!this.error) return '';
if (Array.isArray(this.error?.body)) {
return this.error.body.map(e => e.message).join(', ');
}
return this.error?.body?.message || this.error?.message || 'Unknown error';
}
connectedCallback() {
this.fetchCourses();
}
async fetchCourses() {
this.isLoading = true;
this.error = undefined;
try {
const maxItems = this.normalizeMaxItems(this.maxItems);
const statusValue = this.normalizeStatus(this.statusValue);
const data = await getActiveCourses({ maxItems, statusValue });
this.courses = Array.isArray(data) ? data : [];
} catch (e) {
this.error = e;
this.courses = [];
} finally {
this.isLoading = false;
```

```
}
}
normalizeMaxItems(value) {
const n = Number(value);
if (!Number.isFinite(n) || n <= 0) return 10;
return Math.min(n, 200);
}
normalizeStatus(value) {
const v = (value || '').toString().trim();
return v.length ? v : 'ACTIVE';
}
handleNavigate(event) {
const recordId = event.currentTarget?.dataset?.id;
if (!recordId) return;
this[NavigationMixin.Navigate]({
type: 'standard__recordPage',
attributes: {
recordId,
objectApiName: 'Course__c',
actionName: 'view'
}
});
}
}
```

CourseList.js-meta.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
<apiVersion>60.0</apiVersion>
<isExposed>true</isExposed>
```

```
<masterLabel>Course List</masterLabel>
<targets>
<target>lightningCommunity__Page</target>
<target>lightningCommunity__Default</target>
</targets>
<targetConfigs>
<targetConfig targets="lightningCommunity__Default">
<property name="maxItems" type="Integer" default="10" label="Max items to display"/>
<property name="statusValue" type="String" default="ACTIVE" label="Status API value to filter"/>
</targetConfig>
</targetConfigs>
</LightningComponentBundle>
```

Course registration form

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Student Course Registration - Kongu Engineering College</title>
<style>
body{
font-family: Arial, sans-serif;
background: linear-gradient(135deg,#2aab64,#084f2b);
display:flex;
justify-content:center;
align-items:center;
height:100vh;
margin:0;
```

```
}
.container{
width:900px;
background:white;
border-radius:15px;
display:flex;
overflow:hidden;
box-shadow:0 20px 50px rgba(0,0,0,0.2);
}
.left-panel{
width:40%;
background:#084f2b;
color:white;
padding:40px;
text-align:center;
}
.left-panel img{
width:120px;
margin-bottom:20px;
}
.left-panel h1{
font-size:30px;
margin-bottom:15px;
}
.left-panel p{
line-height:1.6;
font-size:15px;
}
.right-panel{
```

```
width:60%;
padding:40px;
}
h2{
margin-bottom:20px;
color:#084f2b;
}

.form-group{
margin-bottom:18px;
}
label{
display:block;font-weight:bold;
margin-bottom:6px;
}
input, select{
width:100%;
padding:12px;
border-radius:8px;
border:1px solid #ccc;
font-size:14px;
}
button{
padding:12px 25px;
border:none;
background:#2aab64;
color:white;
font-size:16px;
border-radius:8px;
```

```
cursor:pointer;
margin-top:10px;
}button:hover{
background:#084f2b;
}reset{
background:#888;
margin-left:10px;
}.modal{
position:fixed;
top:0;
left:0;
width:100%;
height:100%;
background:rgba(0,0,0,0.5);
display:none;
align-items:center;
justify-content:center;
}
.modal-box{
background:white;
padding:30px;
border-radius:10px;
text-align:center;
width:350px;
}
</style>
</head>
<body>
<div class="container">
```

```
<div class="left-panel">
<imgsrc="..." alt="Kongu Engineering College Logo">
<h1>Student Course Registration</h1>
<p>
Welcome to the Kongu Engineering College Course Registration Portal.
Students can register for available courses through this system.
All registrations are stored in the Salesforce platform for tracking
and monitoring by the placement department.
</p>
</div>
<div class="right-panel">
<h2>Register for a Course</h2>
<form id="courseForm"
action="https://webto.salesforce.com/servlet/servlet.WebToCase?encoding=UTF-8&orgId=00DgL00000KCzLx"
method="POST"
target="hidden_iframe">
<input type="hidden" name="orgid" value="00DgL00000KCzLx">
<input type="hidden" name="retURL" value="http://kongu.ac.in">
<input type="hidden" name="external" value="1">
<div class="form-group">
<label>First Name</label>
<input type="text" name="first_name" placeholder="Enter First Name" required>
</div>
<div class="form-group">
<label>Last Name</label>
<input type="text" name="last_name" placeholder="Enter Last Name" required>
</div>
```

```
<div class="form-group">
<label>Course Name</label>
<select name="course_name" required>
<option value="">Select Course</option>
<option value="Salesforce Administrator">Salesforce Administrator</option>
<option value="Full Stack Development">Full Stack Development</option>
<option value="Data Science">Data Science</option>
<option value="Cloud Computing">Cloud Computing</option>
</select>
</div>
<div class="form-group">
<label>Course Type</label>
<select name="course_type" required>
<option value="">Select Course Type</option>
<option value="Online">Online</option>
<option value="Offline">Offline</option>
<option value="Hybrid">Hybrid</option>
</select>
</div>
<button type="submit">Submit Registration</button>
<button type="reset" class="reset">Clear</button>
</form>
</div>
</div>
<iframe name="hidden_iframe" style="display:none;"></iframe>
<div class="modal" id="successModal">
<div class="modal-box">
<h3>Registration Successful</h3>
<p>
```

```
Your course registration has been submitted successfully.
You will receive confirmation through your registered email.
</p>
<button onclick="closeModal()">OK</button>
</div>
</div>
<script>
const form=document.getElementById("courseForm");
const modal=document.getElementById("successModal");
form.addEventListener("submit",function(){
setTimeout(function(){
modal.style.display="flex";
form.reset();
},800);
});
function closeModal(){
modal.style.display="none";
}
</div>
<button type="submit">Submit Registration</button>
<button type="reset" class="reset">Clear</button>
</form>
</div>
</script>
</body>
</html>
```

B.SCREENSHOTS

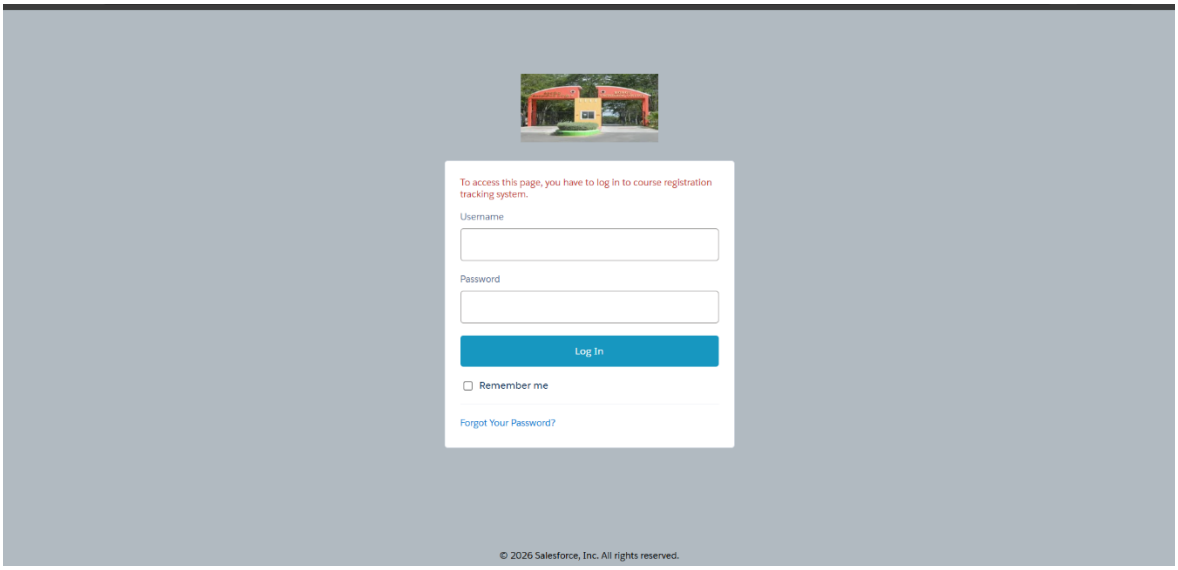


Figure B.1 Login Page

In the FigureB.1 This is the login page for the administrator , placement head and placement faculty .They can enter the login credentials and view their details.

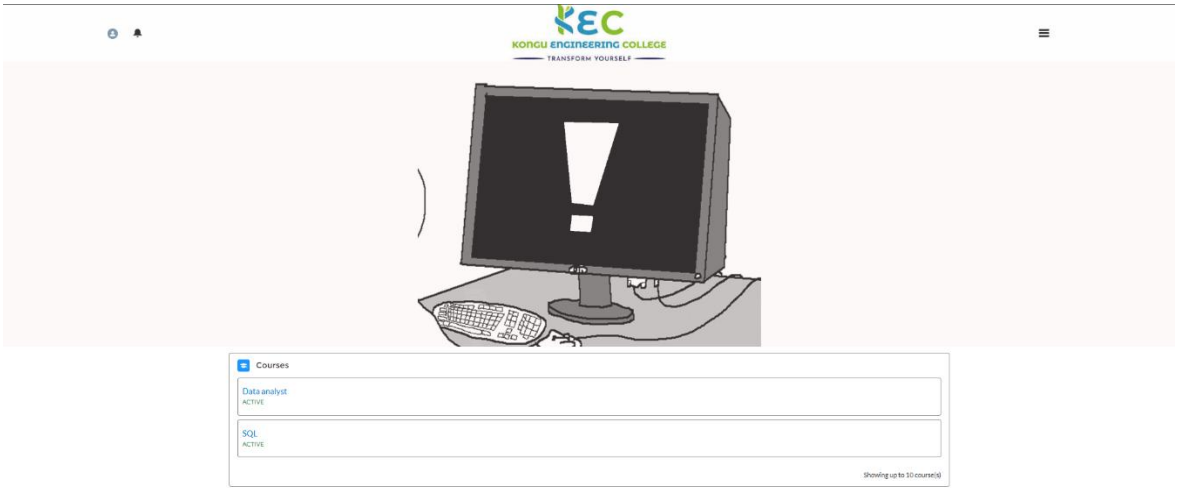
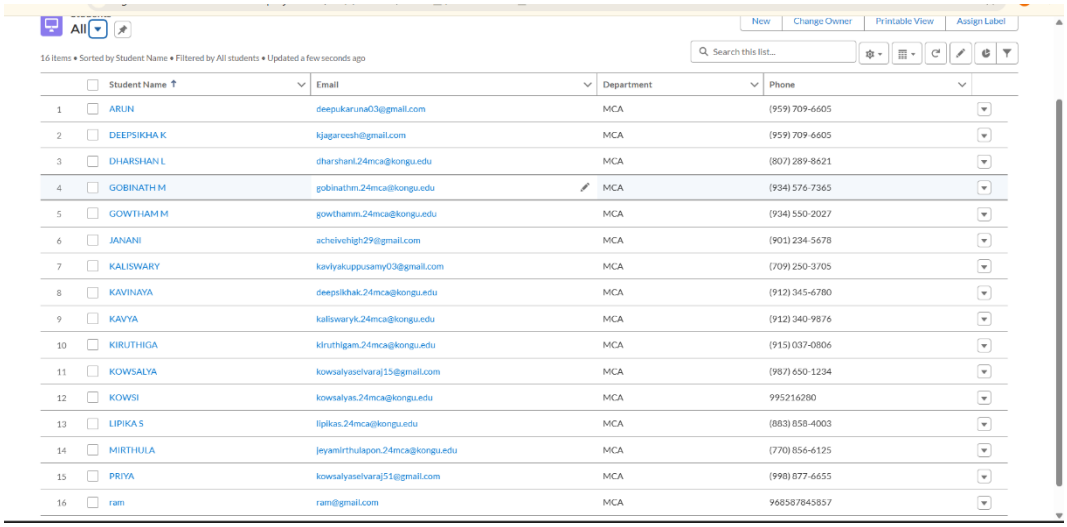


Figure B.2 Home Page

In the FigureB.2 This is the home page for the administrator ,placement head and the placement faculty where the list of active courses will be displayed.



All

New | Change Owner | Printable View | Assign Label

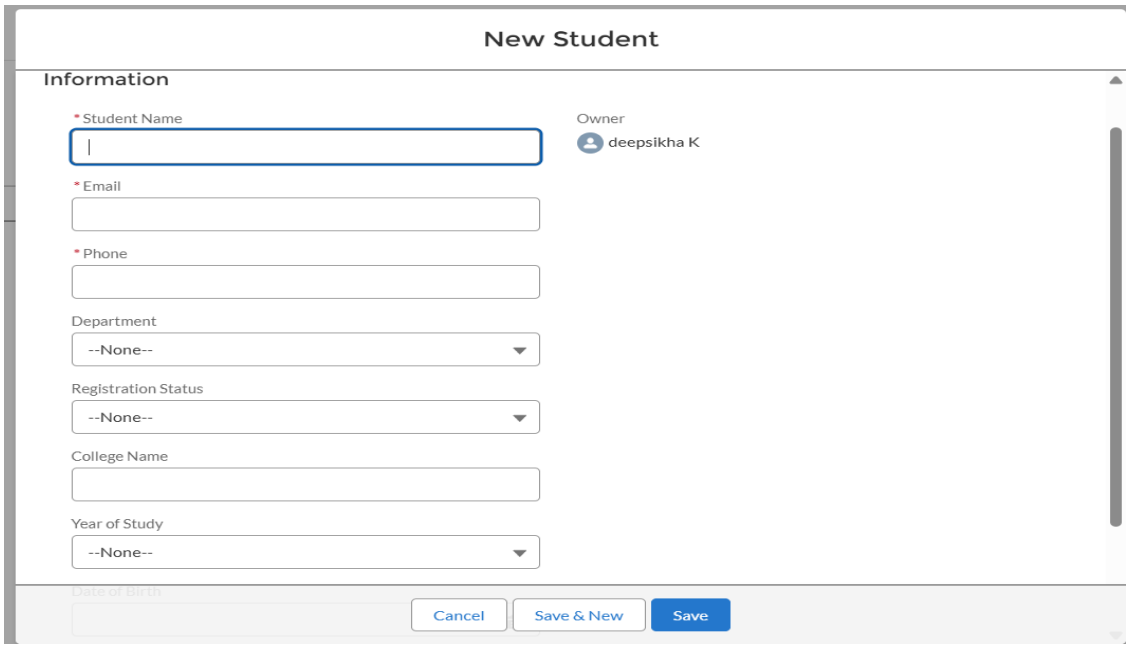
16 items • Sorted by Student Name • Filtered by All students • Updated a few seconds ago

Search this list...

	Student Name	Email	Department	Phone
1	ARUN	deepukaruna03@gmail.com	MCA	(959) 709-6605
2	DEEPSIKHA K	kjagareesh@gmail.com	MCA	(959) 709-6605
3	DHARSHAN L	dharshanl.24mca@kongu.edu	MCA	(807) 289-8621
4	GOBINATH M	gobinathm.24mca@kongu.edu	MCA	(934) 576-7365
5	GOWTHAM M	gowthamm.24mca@kongu.edu	MCA	(934) 550-2027
6	JANANI	acheivehigh29@gmail.com	MCA	(901) 234-5678
7	KALISWARY	kaviyakuppusamy03@gmail.com	MCA	(709) 250-3705
8	KAVINAYA	deepsikhak.24mca@kongu.edu	MCA	(912) 345-6780
9	KAVYA	kaliswaryk.24mca@kongu.edu	MCA	(912) 340-9876
10	KIRUTHIGA	kiruthigam.24mca@kongu.edu	MCA	(915) 037-0806
11	KOWSALYA	kowsalyaselvaraj15@gmail.com	MCA	(987) 650-1234
12	KOWSI	kowsalyas.24mca@kongu.edu	MCA	995216280
13	LIPIKA S	lipikas.24mca@kongu.edu	MCA	(883) 858-4003
14	MIRTHULA	jeyamirthulapon.24mca@kongu.edu	MCA	(770) 856-6125
15	PRIYA	kowsalyaselvaraj51@gmail.com	MCA	(998) 877-6655
16	ram	ram@gmail.com	MCA	968587845857

Figure B.3 Students page

In the Figure B.3 The students who registered will be shown in this list. It contains the details like name,email,phone number ,Department .The student list can be accessed by the administrator but the placement head and the placement faculty can view only the details.



New Student

Information

* Student Name

Owner
deepsikha K

* Email

* Phone

Department
--None--

Registration Status
--None--

College Name

Year of Study
--None--

Date of Birth

Cancel | Save & New | Save

Figure B.4 New Student Registration form

In the Figure B.4 Whenever the students need to register for their course enrolment must fill the form.The students details must be filled by the administrator only.

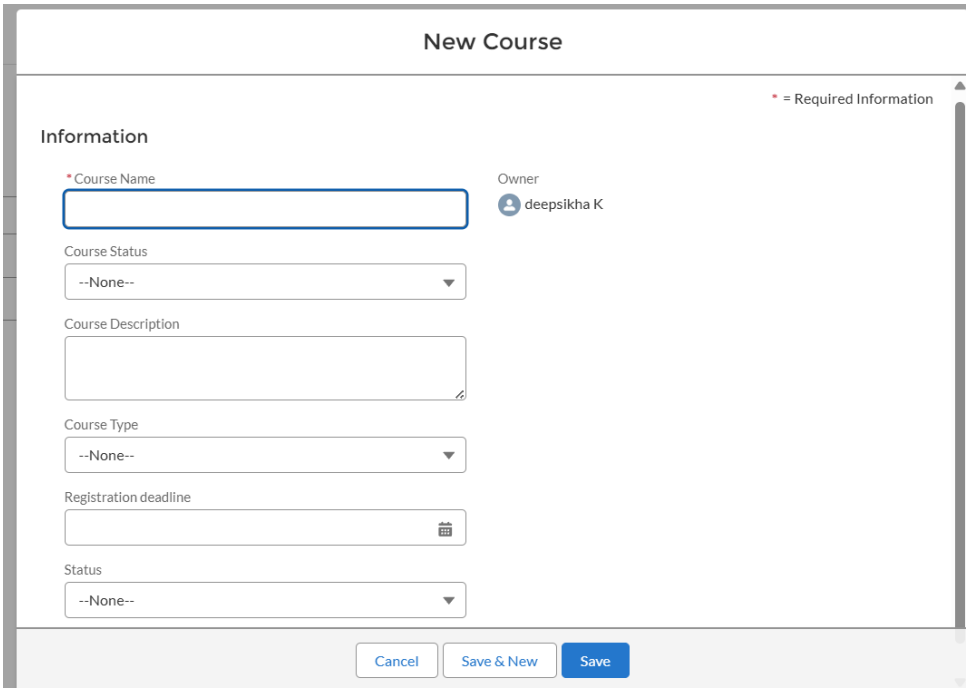


Figure B.5 New Course Registration form

In the Figure B.5 The new courses can be added here by filling the details like course name, status,description etc.., These course can be created by the admin only.

Figure B.6 Course List

In the Figure B.6 The courses created by the admin will be listed here. From this page we can see what are the courses are created by the admin.

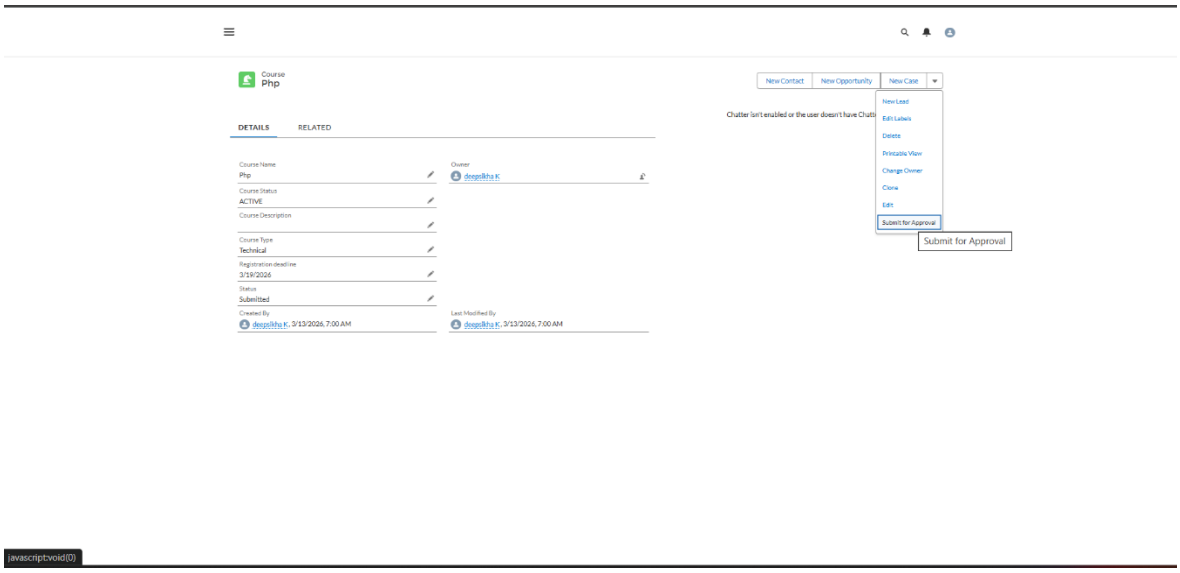


Figure B.7 Submit for approval

In the FigureB.7 Whenever the new courses are created the admin will give submit for approval request to the Placement faculty.

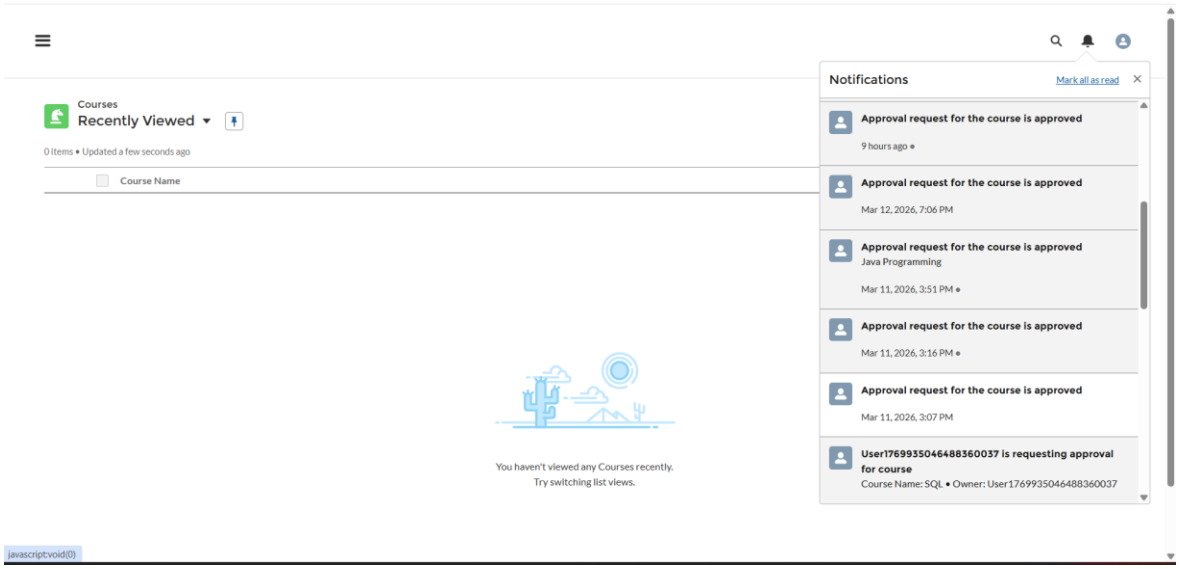


Figure B.8 Approval notifications

In this figure B.8 When the approval requests reaches the placement faculty they will receive the notification then the placement faculty will either approve or reject.

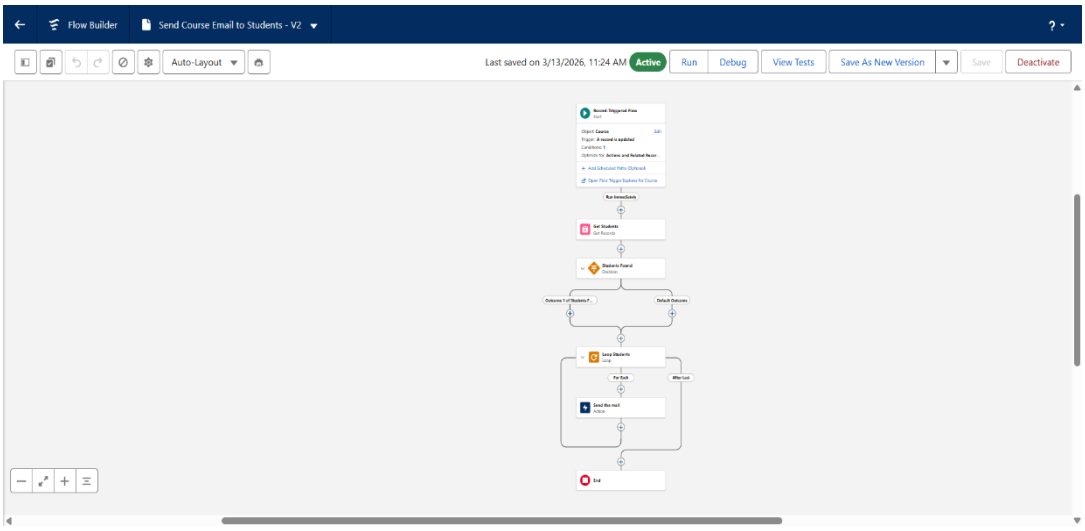


Figure B.9 Flow to send email

In the FigureB.9 This flow will send the mail to the students to register for their courses. They will the retrieve the students to send the mail to the students.

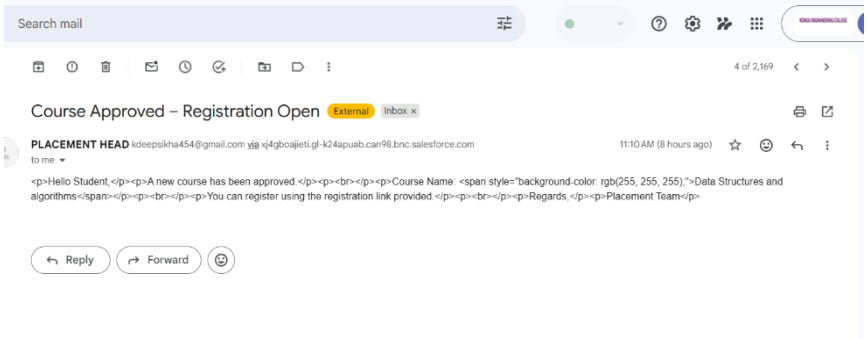


Figure B.10 Mail Received

In the Figure B.10 The students will receive the mail for registering their courses .By the link we can register for the courses. The mail will be sent to the students who are registered

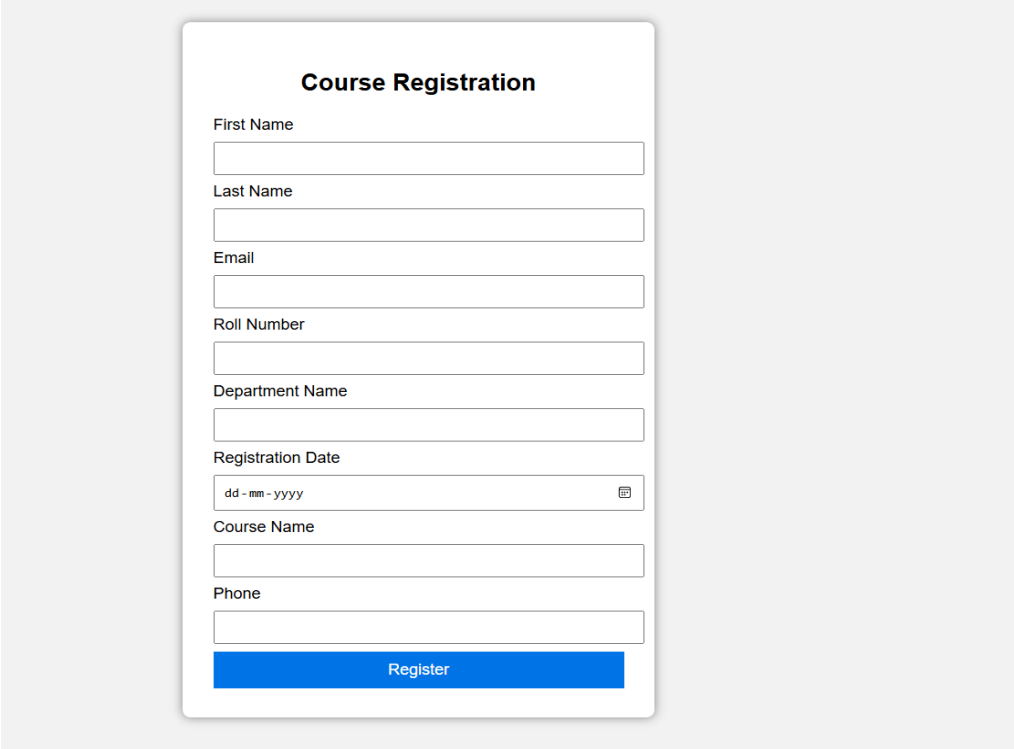


Figure B.11 Students course registration form

In the Figure B.11The students can register for their course by filling the essential details like First name , Last Name , Email ,Roll Number etc.., Then these details will be stored in a cloud.

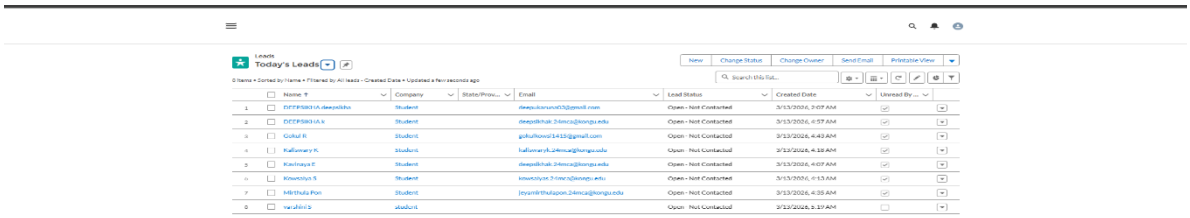
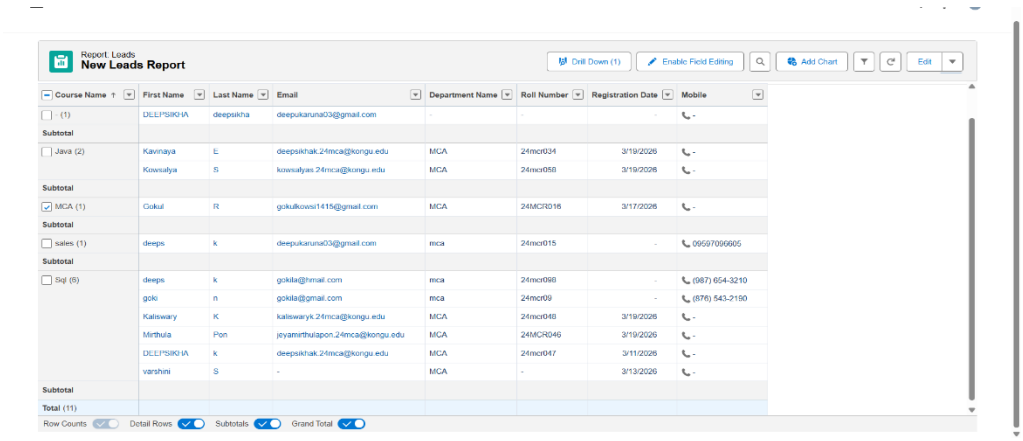


Figure B.12 No.of.students registered

In the Figure B.12 this page we can see the no.of students registered for their courses.This can be sorted by todays lead ,unread leads.



Report: Leads
New Leads Report

Drill Down (1) | Enable Field Editing | Add Chart | Edit

Course Name	First Name	Last Name	Email	Department Name	Roll Number	Registration Date	Mobile
(1)	DEEPSIKHA	deepsikha	deepukaruna03@gmail.com	-	-	-	-
Subtotal							
Java (2)	Kavinaya	E	deepsikhak.24mca@kongu.edu	MCA	24mcr034	3/19/2026	-
	Kowsalya	S	kowsalyas.24mca@kongu.edu	MCA	24mcr058	3/19/2026	-
Subtotal							
MCA (1)	Gokul	R	gokulkowsi1415@gmail.com	MCA	24MCR016	3/17/2026	-
Subtotal							
sales (1)	deeps	k	deepukaruna03@gmail.com	mca	24mcr015	-	09597096605
Subtotal							
Sql (6)	deeps	k	gokila@hmail.com	mca	24mcr098	-	(987) 654-3210
	goki	n	gokila@gmail.com	mca	24mcr09	-	(876) 543-2180
	Kaliswary	K	kaliswaryk.24mca@kongu.edu	MCA	24mcr048	3/19/2026	-
	Mirthula	Pon	joyamirthulapon.24mca@kongu.edu	MCA	24MCR046	3/19/2026	-
	DEEPSIKHA	k	deepsikhak.24mca@kongu.edu	MCA	24mcr047	3/11/2026	-
	varshini	S	-	MCA	-	3/13/2026	-
Subtotal							
Total (11)							

Row Counts | Detail Rows | Subtotals | Grand Total

Figure B.13 Leads report

In this Figure B.13 The lead report will be created by the admin .In this report we can see the name of the students registered , email , course name .These report will be shared to the placement faculty.

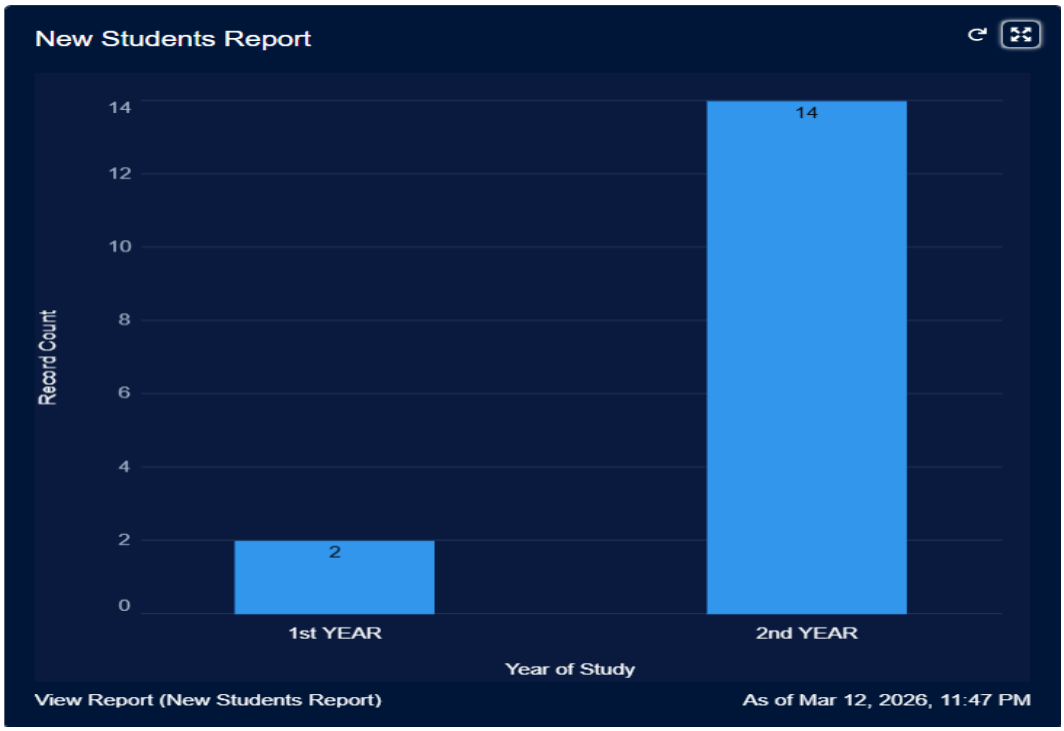


Figure B.14 Students report

In this Figure B.14 The report will be created by the admin and it is shared to the placement faculty then the placement faculty will subscribe the report to the placement head.

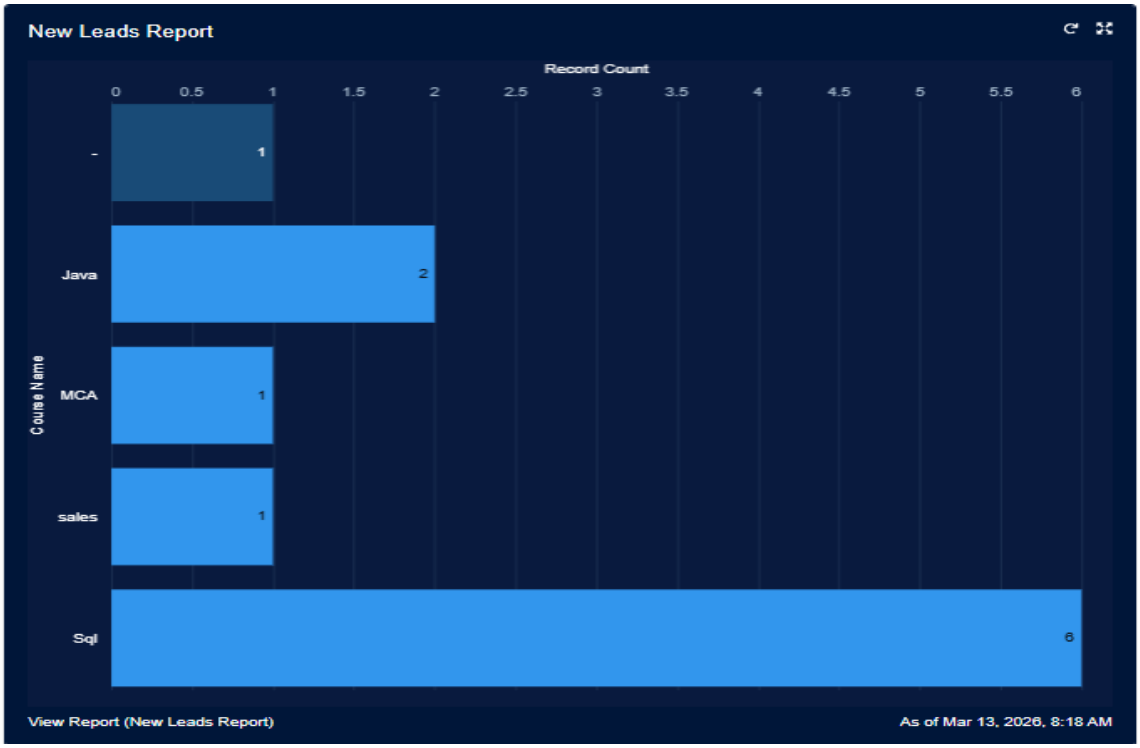


Figure B.15 Registration count

In this Figure B.15 we can see the no.of registrations count for each courses and it displays the course wise with the registered count.

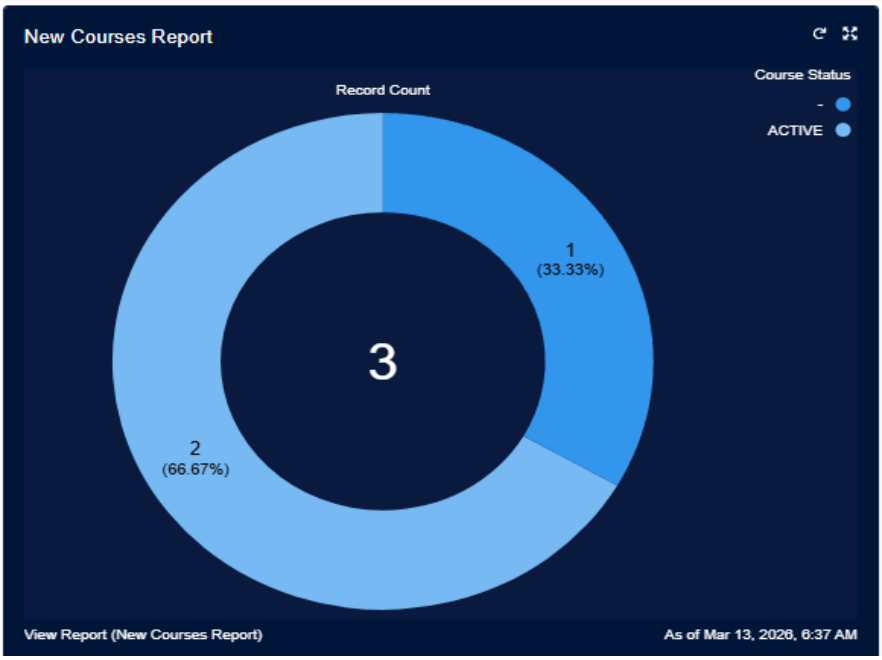


Figure B.16 Course Dashboard

In this Figure B.16 The dashboard we can easily figure out the no.of.students registered for individual courses and the count has been shown.

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10. https://compliance.salesforce.com/en/documents
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SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG 4 – Quality Education

- The system helps students easily access and register for skill development courses offered by the institution. This improves learning opportunities and supports continuous education for students.
- It provides a digital platform where course information is organized and easily accessible to students. This makes it easier for students to explore and participate in different training programs.
- The system supports the management of educational programs by allowing administrators to create and update course details. This improves the efficiency of managing academic and training activities.

SDG 9 – Industry, Innovation and Infrastructure

- The project is developed using the Salesforce cloud platform, which represents the use of modern technology in education. This supports innovation and digital transformation in educational institutions.
- The system automates processes such as course creation, approval and registration.This reduces manual work and improves the efficiency of institutional infrastructure.
- By implementing a cloud-based system, the project supports the development of reliable and scalable digital infrastructure. This helps institutions manage large amounts of data effectively.

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- The system allows students to register for training courses that enhance their knowledge and technical skills. These skills help students become more prepared for employment opportunities.
- It supports placement training activities by providing a platform to manage and monitor different career-oriented courses. This contributes to student career development and professional growth.
- The platform helps track student participation in skill development programs. This ensures that students are actively involved in improving their career prospects.